

SGA 3, Exposé XXIII
Reductive Groups: Uniqueness of Pinned Groups
English Loop-1 Reconstruction

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Contents

Exposé XXIII. Uniqueness of Pinned Groups	1
1. Pinnings	1
1.7. The pinned groups Z_{Δ_1}	3
1.8. Study of the group $\text{Norm}_G(T)$	4
1.9. Faithfulness of the functor $\mathcal{R}$	5
2. Generators and relations for a pinned group	6
2.1.1. Proof	6
3. Groups of Semisimple Rank 2	14
3.1. Generalities	14
3.2. Groups of Type A_2	15
3.3. Groups of Type B_2	17
3.4. Groups of Type G_2	20
3.5. Explicit Form of the Generators-and-Relations Theorem	23
4. Uniqueness of Pinned Groups: Fundamental Theorem	25
5. Corollaries of the Fundamental Theorem	29
6. Chevalley Systems	31

Exposé XXIII

Reductive Groups: Uniqueness of Pinned Groups

by *M. Demazure*

¹ The aim of this Exposé is to prove the uniqueness theorem (Theorem 4.1). It was proved by Chevalley in the case of an algebraically closed field; the reduction-to-rank-two method used here is likewise due to Chevalley (see *Bible*, Exposés 23 and 24). Along the way, we obtain an explicit description of reductive groups by generators and relations (3.5).

1. Pinnings

Definition 1.1.— *Let S be a scheme and let (G, T, M, R) be an S -split group (XXII, 1.13). A pinning ² of this split group is the datum of a system Δ of simple roots of R , together with, for each $\alpha \in \Delta$, a section*

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times.$$

In other words, a pinning of the reductive group G over the nonempty scheme S is the datum of:

- (i) a maximal torus T ;
- (ii) an abelian group M and an isomorphism $T \simeq D_S(M)$;
- (iii) a system of roots R of G with respect to T ;
- (iv) a system Δ of simple roots of R ;
- (v) an element $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$ for every $\alpha \in \Delta$, that is, an element

$$u_\alpha = \exp_\alpha(X_\alpha) \in U_\alpha^\times(S) \quad \text{for every } \alpha \in \Delta,$$

¹*Editors' note.* Version of 13 October 2024.

²*Editors' note.* Demazure tells us that behind this terminology lies the image of the butterfly (supplied to him by Grothendieck): the body is a maximal torus T , the wings are two Borel subgroups opposite with respect to T ; one spreads the butterfly by laying out the wings and then fixes elements in the additive groups (pins) to rigidify the situation, that is, to eliminate the automorphisms.

satisfying condition (D1) of Exposé XXII, 1.13 (indeed, condition (D2) of *loc. cit.* is automatic³).

Every split group admits a pinning; in particular, every reductive group is locally pinnable for the étale topology.

1.2.— If G is a pinned S -group, that is, an S -split group equipped with a pinning, it is canonically endowed with the system of positive roots R^+ defined by Δ , the corresponding Borel subgroup $B = B_{R^+}$, the opposite Borel subgroup $B^- = B_{R^-}$, the unipotent groups $U = B_u$ and $U^- = (B^-)_u$, the open subscheme $U^- \cdot T \cdot U$, and so forth. Likewise, for each $\alpha \in \Delta$, there is a canonical isomorphism of vector groups

$$p_\alpha : \mathbb{G}_{a,S} \xrightarrow{\sim} U_\alpha, \quad x \mapsto \exp_\alpha(xX_\alpha) = u_\alpha^x,$$

normalized by T with multiplier α , and whose datum is equivalent to that of X_α (Exposé XXII, 1.1).

By duality one obtains an $X_{-\alpha} \in \Gamma(S, \mathfrak{g}^{-\alpha})^\times$ and an isomorphism

$$p_{-\alpha} : \mathbb{G}_{a,S} \xrightarrow{\sim} U_{-\alpha}$$

which is contragredient to the preceding one (Exposé XXII, 1.3). We set (Exposé XX, 3.1)

$$w_\alpha = w_\alpha(X_\alpha) = p_\alpha(1)p_{-\alpha}(-1)p_\alpha(1) = p_{-\alpha}(-1)p_\alpha(1)p_{-\alpha}(-1).$$

Then (*loc. cit.*, 3.1 and 3.7)

$$w_\alpha^2 = \alpha^*(-1), \quad \text{int}(w_\alpha)t = s_\alpha(t) = t \cdot \alpha^*(\alpha(t)^{-1}),$$

$$\begin{aligned} \text{int}(w_\alpha)p_\alpha(x) &= p_{-\alpha}(-x) = p_{-\alpha}(x)^{-1}, & \text{Ad}(w_\alpha)X_\alpha &= -X_{-\alpha}, \\ \text{int}(w_\alpha)p_{-\alpha}(x) &= p_\alpha(-x) = p_\alpha(x)^{-1}, & \text{Ad}(w_\alpha)X_{-\alpha} &= -X_\alpha. \end{aligned}$$

We shall systematically use the preceding notation in what follows.

Definition 1.3.— Let S be a scheme, and suppose that

$$(G, T, M, R, \Delta, (X_\alpha)) \quad \text{and} \quad (G', T', M', R', \Delta', (X'_{\alpha'}))$$

are two pinned S -groups. The morphism of split groups (Exposé XXII, 4.2.1)

$$f : G \longrightarrow G'$$

is said to be compatible with the pinnings, or to define a morphism of pinned groups, if the associated bijection $d : R \rightarrow R'$ (cf. *loc. cit.*) satisfies $d(\Delta) = \Delta'$ and if, for every $\alpha \in \Delta$,

$$f(\exp_\alpha(X_\alpha)) = \exp_{d(\alpha)}(X'_{d(\alpha)}), \quad \text{i.e.} \quad f(u_\alpha) = u'_{d(\alpha)}.$$

4

1.4.— If $q(\alpha)$ denotes the integer of *loc. cit.*, then

$$f(p_\alpha(x)) = p'_{d(\alpha)}(x^{q(\alpha)}) \quad (\alpha \in \Delta),$$

³*Editors' note.* It is implied by condition (v), namely the existence of a section $X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times$.

⁴*Editors' note.* The bijection $R \xrightarrow{\sim} R'$ is denoted by d (rather than by u) in order to avoid the notation $u'_{u(\alpha)}$.

and hence

$$f(p_{-\alpha}(x)) = p'_{-d(\alpha)}(x^{q(\alpha)}), \quad f(w_\alpha) = w'_{d(\alpha)}.$$

Recall (Exposé XXII, 4.2) that, for every $\alpha \in R$, every $z \in \mathbb{G}_m(S')$, and every $t \in T(S')$,

$$f(\alpha^*(z)) = d(\alpha)^*(z)^{q(\alpha)} = d(\alpha)^*(z^{q(\alpha)}), \quad d(\alpha)(f(t)) = \alpha(t)^{q(\alpha)}.$$

1.5.— We call a root datum equipped with a system of simple roots a *pinned root datum*, and a p -morphism of root data (Exposé XXI, 6.8) that sends simple roots to simple roots a *p -morphism of pinned root data*.

If G is a pinned S -group, we denote by $\mathcal{R}(G)$ the corresponding pinned root datum (the root datum of Exposé XXII, 1.14 equipped with Δ). Let p be the integer defined in Exposé XXII, 4.2.2. Then:

Scholium 1.6.— *The correspondence $G \mapsto \mathcal{R}(G)$ defines a functor from the category of pinned reductive S -groups to the category of pinned root data (whose morphisms are the p -morphisms).*

1.7. The pinned groups Z_{Δ_1}

Let Δ_1 be a subset of the system Δ of simple roots of the pinned group G . Let T_{Δ_1} be the maximal torus of $\bigcap_{\alpha \in \Delta_1} \ker(\alpha)$, and set

$$Z_{\Delta_1} = \text{Centr}_G(T_{\Delta_1}).$$

Put $R' = \mathbb{Z}\Delta_1 \cap R$. It is known (Exposé XXII, 5.10.7) that Z_{Δ_1} is a reductive S -group with radical T_{Δ_1} , that (T, M, R') is a splitting of this group, and that Δ_1 is a system of simple roots. Thus

$$(Z_{\Delta_1}, T, M, R', \Delta_1, (X_\alpha)_{\alpha \in \Delta_1})$$

is a pinned S -group. We shall always give Z_{Δ_1} this pinning. In particular, we shall consider

$$Z_\alpha = Z_{\{\alpha\}}, \quad Z_{\alpha\beta} = Z_{\{\alpha, \beta\}}.$$

Write $B_{\Delta_1} = B \cap Z_{\Delta_1}$. It is known (*loc. cit.*) that this is the canonical Borel subgroup⁵ of Z_{Δ_1} , and that its unipotent part is $U_{\Delta_1} = U \cap Z_{\Delta_1}$. In particular,

$$B_\alpha = T \cdot U_\alpha.$$

We write

$$U_{\alpha\beta} = U_{\{\alpha, \beta\}} = U \cap Z_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma,$$

where $R_{\alpha\beta}^+$ is the set of positive roots that are linear combinations of α and β .

Now let $f : G \rightarrow G'$ be a morphism of pinned S -groups. If $d : R \rightarrow R'$ is the corresponding bijection and $\Delta_1 \subset \Delta$, then $d(\Delta_1) = \Delta'_1 \subset \Delta'$. Clearly f maps T_{Δ_1} into $T'_{\Delta'_1}$, and hence Z_{Δ_1} into $Z'_{\Delta'_1}$. The corresponding S -morphism

$$f_{\Delta_1} : Z_{\Delta_1} \longrightarrow Z'_{\Delta'_1}$$

is compatible with the canonical pinnings. It defines a morphism of pinned root data

$$\mathcal{R}(f_{\Delta_1}) : \mathcal{R}(Z_{\Delta_1}, T, M, \dots) \longrightarrow \mathcal{R}(Z'_{\Delta'_1}, T', M', \dots),$$

and the underlying morphisms $M' \rightarrow M$ of $\mathcal{R}(f)$ and $\mathcal{R}(f_{\Delta_1})$ coincide.

⁵*Editors' note.* That is, the Borel subgroup of Z_{Δ_1} corresponding to $R'^+ = R' \cap R^+$.

1.8. Study of the group $\text{Norm}_G(T)$

For every pair (α, β) of simple roots, let $n_{\alpha\beta}$ be the order of $s_\alpha s_\beta$ in the Weyl group W . In particular $n_{\alpha\alpha} = 1$, and therefore $(w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S)$.

Definition 1.8.1.— For every $(\alpha, \beta) \in \Delta \times \Delta$, set

$$t_{\alpha\beta} = (w_\alpha w_\beta)^{n_{\alpha\beta}} \in T(S).$$

Moreover, set (Exposé XX, 3.1)

$$t_\alpha = t_{\alpha\alpha} = w_\alpha^2 = \alpha^*(-1) \in T(S).$$

Proposition 1.8.2.— Let H be an S -functor in groups that carries direct sums of schemes to products (for example, a sheaf for the Zariski topology). Let

$$f_T : T \longrightarrow H$$

be a morphism of groups, and let $h_\alpha \in H(S)$ ($\alpha \in \Delta$). There exists a necessarily unique morphism of groups

$$f_N : \text{Norm}_G(T) \longrightarrow H$$

inducing f_T on T and satisfying $f_N(w_\alpha) = h_\alpha$ for $\alpha \in \Delta$ if and only if the following two conditions hold:

(i) for every $\alpha \in \Delta$,

$$f_T(s_\alpha(t)) = h_\alpha f_T(t) h_\alpha^{-1}$$

for every $t \in T(S')$, $S' \rightarrow S$; equivalently, $f_T \circ s_\alpha = \text{int}(h_\alpha) \circ f_T$;

(ii) for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$f_T(t_{\alpha\beta}) = (h_\alpha h_\beta)^{n_{\alpha\beta}}.$$

Proof. Equip (Sch) with the topology $\mathcal{T}$ generated by the pretopology whose covering families are direct sums. The hypothesis says that H is a $\mathcal{T}$ -sheaf. Let L be the free group on the generators $(m_\alpha)_{\alpha \in R}$, and let L_1 be the invariant subgroup generated by

$$(m_\alpha m_\beta)^{n_{\alpha\beta}}, \quad (\alpha, \beta) \in \Delta \times \Delta.$$

Let $g : L \rightarrow W$ be defined by $g(m_\alpha) = s_\alpha$. It is known (Exposé XXI, 5.1) that g induces an isomorphism $\bar{g} : L/L_1 \xrightarrow{\sim} W$. Let L act on T through g , equivalently by $m_\alpha \cdot t = s_\alpha(t)$. Writing L_S for the constant group defined by L , form the semidirect product $N = T \cdot L_S$. There is a unique morphism of S -groups

$$h : N = T \cdot L_S \longrightarrow \text{Norm}_G(T)$$

such that $h(m_\alpha) = w_\alpha$ and $h(t) = t$ for every $t \in T(S')$, $S' \rightarrow S$.

Let N_1 be the invariant subgroup sheaf of N generated by

$$t_{\alpha\beta}^{-1} (m_\alpha m_\beta)^{n_{\alpha\beta}}, \quad (\alpha, \beta) \in \Delta \times \Delta.$$

Clearly $N_1 \subset \ker h$, so h induces

$$h_1 : N/N_1 \longrightarrow \text{Norm}_G(T).$$

We prove that h_1 is an isomorphism. Since $h|_T$ is the canonical immersion, hence a monomorphism, the canonical morphism $T \rightarrow N/N_1$ is also a monomorphism and induces an isomorphism $T \xrightarrow{\sim} TN_1/N_1$. Thus h_1 induces an isomorphism $TN_1/N_1 \xrightarrow{\sim} T$. It remains only to show that

$$h_2 : N/TN_1 \longrightarrow \text{Norm}_G(T)/T$$

is an isomorphism. But TN_1 is the invariant $\mathcal{T}$ -subgroup sheaf of N generated by T and the $(m_\alpha m_\beta)^{n_{\alpha\beta}}$; hence it is the subgroup sheaf generated by T and L_1 , namely $T \cdot (L_1)_S$. Therefore h_2 identifies with

$$\bar{g} : L_S/(L_1)_S \longrightarrow W_S,$$

which is an isomorphism by construction.

The proof of the proposition is now immediate. Necessity is clear. Conversely, condition (i) gives a morphism

$$u : N \longrightarrow H$$

such that $u(m_\alpha) = h_\alpha$ for $\alpha \in \Delta$ and $u|_T = f_T$. Condition (ii) says that u vanishes on N_1 , and the result follows.

1.9. Faithfulness of the functor $\mathcal{R}$

Proposition 1.9.1.— *The functor $\mathcal{R}$ of 1.6 is faithful: if*

$$f, g : G \rightrightarrows G'$$

are two morphisms of pinned groups such that $\mathcal{R}(f) = \mathcal{R}(g)$, then $f = g$.

Indeed, f and g agree on T , on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$); it therefore suffices to apply:

Lemma 1.9.2.— *Let S be a scheme, let G be a pinned reductive S -group, and let H be an S -presheaf in groups separated for the fppf topology. Let*

$$f, g : G \rightrightarrows H$$

be two morphisms of S -groups. The following conditions are equivalent:

- (i) $f = g$;
- (ii) f and g agree on T , on every U_α ($\alpha \in \Delta$), and on every $U_{-\alpha}$ ($\alpha \in \Delta$);
- (iii) f and g agree on T and on every U_α ($\alpha \in \Delta$), and $f(w_\alpha) = g(w_\alpha)$ for every $\alpha \in \Delta$.

Proof. The implication (i) $\Rightarrow$ (ii) is trivial, while (ii) $\Rightarrow$ (iii) follows immediately from the definition of w_α in 1.2. It remains to prove (iii) $\Rightarrow$ (i). If $\alpha \in R$, there is a sequence $\{\alpha_i\} \subset \Delta$ such that

$$\alpha = s_{\alpha_1} s_{\alpha_2} \cdots s_{\alpha_n}(\alpha_{n+1}),$$

and hence

$$U_\alpha = \text{int}(w_{\alpha_1} \cdots w_{\alpha_n})U_{\alpha_{n+1}}.$$

Thus f and g agree on every U_α . They consequently agree on Ω , and therefore agree everywhere (Exposé XXII, 4.1.11).

Remark 1.9.3.— *If G is semisimple, the hypothesis that f and g agree on T may be omitted in (ii) and (iii). Indeed, G , as an fppf sheaf, is generated by the U_α , $\alpha \in R$ (Exposé XXII, 6.2.2(a)).*

2. Generators and relations for a pinned group

Throughout this section G is a fixed pinned S -group. For $\alpha, \beta \in \Delta$, we use the notation $Z_\alpha, Z_{\alpha\beta}, U_{\alpha\beta}, R_{\alpha\beta}^+$ of 1.7.

Theorem 2.1.— *Let S be a scheme, let G be a pinned S -group, and let H be an fppf S -sheaf in groups. Suppose given morphisms of groups*

$$f_N : \text{Norm}_G(T) \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in R).$$

There exists a necessarily unique morphism of groups

$$f : G \longrightarrow H$$

extending f_N and all the f_α , $\alpha \in R$, if and only if the following conditions hold:

(i) *for every $\alpha \in \Delta$ and $\beta \in R$,*

$$\text{int}(f_N(w_\alpha)) \circ f_\beta = f_{s_\alpha(\beta)} \circ \text{int}(w_\alpha);$$

(ii) *for every $\alpha \in \Delta$, there exists a morphism of groups*

$$F_\alpha : Z_\alpha \longrightarrow H$$

extending f_α , $f_{-\alpha}$, and $f_N|_{\text{Norm}_{Z_\alpha}(T)}$;

(iii) *for every $(\alpha, \beta) \in \Delta \times \Delta$, there exists a morphism of groups $U_{\alpha\beta} \rightarrow H$ inducing f_γ on U_γ for every $\gamma \in R_{\alpha\beta}^+$ (so $U_\gamma \subset U_{\alpha\beta}$).*

2.1.1. Proof

The conditions are plainly necessary. Choose a totally ordered group structure on $\Gamma_0(R)$ for which the positive elements of R are precisely R^+ (Exposé XXI, 3.5.6); every product indexed by a subset of R will be taken in this order. Let f_T denote the restriction of f_N to T , and define a morphism of sets

$$f : \Omega \longrightarrow H$$

by

$$f \left(\prod_{\alpha \in R^-} y_\alpha \cdot t \cdot \prod_{\alpha \in R^+} x_\alpha \right) = \prod_{\alpha \in R^-} f_\alpha(y_\alpha) \cdot f_T(t) \cdot \prod_{\alpha \in R^+} f_\alpha(x_\alpha).$$

Every morphism satisfying the theorem must extend this map. Conversely, any morphism of groups $f : G \rightarrow H$ extending it also extends f_N : since it extends f_α and $f_{-\alpha}$, one has

$$f(w_\alpha) = F_\alpha(w_\alpha) = f_N(w_\alpha),$$

and it extends f_T by hypothesis. By Exposé XXII, 4.1.11(ii), it therefore remains to prove:

Proposition 2.1.2.— *The morphism $f : \Omega \rightarrow H$ defined above is generically multiplicative: for every $S' \rightarrow S$ and every $x, y \in \Omega(S')$ such that $xy \in \Omega(S')$, one has $f(xy) = f(x)f(y)$.*

Lemma 2.1.3.— *Let $n \in \text{Norm}_G(T)(S')$, $S' \rightarrow S$, and let $\alpha, \beta \in R$ satisfy $\text{int}(n)U_\alpha = U_\beta$, equivalently $n(\alpha) = \beta$. Then*

$$\text{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \text{int}(n).$$

It suffices to verify this formula for a system of generators of $\text{Norm}_G(T)$. It holds for every w_α , $\alpha \in \Delta$, by 2.1(i), so it remains to treat $n \in T(S')$. This is immediate from 2.1(ii) when α is simple. Otherwise choose $w \in W$ with $w^{-1}(\alpha) \in \Delta$. Writing w as a product of simple reflections⁶, one reduces to showing that if the formula holds for α and every n , it also holds for $w_{\alpha_0}(\alpha)$ and $t \in T(S')$, where $\alpha_0 \in \Delta$. By 2.1(i),

$$\begin{aligned} \text{int}(f_N(t)) \circ f_{w_{\alpha_0}(\alpha)} &= \text{int}(f_N(tw_{\alpha_0})) \circ f_\alpha \circ \text{int}(w_{\alpha_0}^{-1}) \\ &= f_{w_{\alpha_0}(\alpha)} \circ \text{int}(tw_{\alpha_0}) \circ \text{int}(w_{\alpha_0}^{-1}). \end{aligned}$$

Lemma 2.1.4.— *The restriction of f to U (respectively U^-) is a morphism of groups. In particular, this restriction is independent of the chosen ordering of the roots.*

It suffices to prove the assertion for U . By Exposé XXII, 5.5.8, it is enough to show that, for every pair $\alpha < \beta$ of positive roots and every $x_\alpha \in U_\alpha(S')$, $x_\beta \in U_\beta(S')$, $S' \rightarrow S$,

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = f(x_\beta x_\alpha x_\beta^{-1}).$$

By Exposé XXII, 5.5.2, there are elements $x_\gamma \in U_\gamma(S')$, where $\gamma = i\alpha + j\beta \in R$, $i > 0$, $j \geq 0$ ⁷, such that

$$x_\beta x_\alpha x_\beta^{-1} = \prod_{\gamma} x_\gamma.$$

It remains to verify

$$f_\beta(x_\beta)f_\alpha(x_\alpha)f_\beta(x_\beta^{-1}) = \prod_{\substack{\gamma=i\alpha+j\beta \\ i>0, j\geq 0}} f_\gamma(x_\gamma).$$

By Exposé XXI, 3.5.4, there is a $w \in W$ such that $w(\alpha) = \alpha_0 \in \Delta$ and $w(\beta) \in A_{\alpha_0\beta_0}$, in the notation of 1.7, for some $\beta_0 \in \Delta$. Lift w to $n \in \text{Norm}_G(T)(S)$ by Exposé XXII, 3.8. It suffices to verify the relation after conjugation by $f_N(n)$; by 2.1.3, this reduces to $\alpha, \beta \in R_{\alpha_0\beta_0}^+$, where condition 2.1(iii) applies.

Lemma 2.1.5.— *Let $n \in \text{Norm}_G(T)(S)$ satisfy $\text{int}(n)U = U^-$, so that n represents the symmetry of the Weyl group⁸ (Exposé XXI, 3.6.14). For every $u \in U(S')$, $S' \rightarrow S$ (respectively $u \in U^-(S')$), one has*

$$f(nun^{-1}) = f_N(n)f(u)f_N(n^{-1}).$$

This follows immediately from 2.1.3 and 2.1.4.

Lemma 2.1.6.— *Let $u \in B(S')$, $v \in B^-(S')$, and $g \in \Omega(S')$, where $S' \rightarrow S$. Then*

$$f(vgu) = f(v)f(g)f(u).$$

Write

$$v = v_1 t_1, \quad g = v_2 t_2 u_2, \quad u = t_3 u_3,$$

⁶Editors' note. "Fundamental symmetries" was replaced by "simple reflections."

⁷Editors' note. The printed $i, j > 0$ was corrected to $i > 0, j \geq 0$, since x_α occurs in the product on the right.

⁸Editors' note. Relative to Δ .

with $v_i \in U^-(S')$, $t_i \in T(S')$, and $u_i \in U(S')$. On the one hand,

$$f(v)f(g)f(u) = f(v_1)f_T(t_1)f(v_2)f_T(t_2)f(u_2)f_T(t_3)f(u_3).$$

On the other hand,

$$\begin{aligned} f(vgu) &= f(v_1t_1v_2t_1^{-1}t_1t_2t_3t_3^{-1}u_2t_3u_3) \\ &= f(v_1 \cdot t_1v_2t_1^{-1})f_T(t_1t_2t_3)f(t_3^{-1}u_2t_3 \cdot u_3). \end{aligned}$$

Using 2.1.4 on the two outer factors gives

$$f(vgu) = f(v_1)f(t_1v_2t_1^{-1})f_T(t_1t_2t_3)f(t_3^{-1}u_2t_3)f(u_3).$$

It remains only to use

$$\begin{aligned} f(t_1v_2t_1^{-1}) &= f_T(t_1)f(v_2)f_T(t_1)^{-1}, \\ f(t_3^{-1}u_2t_3) &= f_T(t_3)^{-1}f(u_2)f_T(t_3), \end{aligned}$$

which follow from the definition of f and 2.1.3.

Lemma 2.1.7.— *Let $\alpha \in \Delta$ and*

$$g \in \Omega(S') \cap \text{int}(w_\alpha)^{-1}\Omega(S'), \quad S' \rightarrow S.$$

Then

$$f(w_\alpha g w_\alpha^{-1}) = f_N(w_\alpha)f(g)f_N(w_\alpha)^{-1}.$$

By Exposé XXII, 5.6.8, write

$$g = ax_{-\alpha}tx_\alpha b,$$

where $a \in U_{-\hat{\alpha}}^-(S')$, $x_{-\alpha} \in U_{-\alpha}(S')$, $t \in T(S')$, $x_\alpha \in U_\alpha(S')$, and $b \in U_{\hat{\alpha}}(S')$. By 2.1.3 and 2.1.4, and because s_α permutes the positive roots other than α (Exposé XXI, 3.3.2),

$$\text{int}(w_\alpha)a \in U_{-\hat{\alpha}}^-(S'), \quad \text{int}(w_\alpha)b \in U_{\hat{\alpha}}(S').$$

The desired formula holds for $g = a$ and $g = b$. By 2.1.6 it therefore suffices to treat $g = x_{-\alpha}tx_\alpha \in Z_\alpha(S')$; in that case everything takes place in Z_α , and condition 2.1(ii) applies.

Lemma 2.1.8.— *Let $n \in \text{Norm}_G(T)(S)$. For every $S' \rightarrow S$ and every $g \in \Omega(S')$ such that $\text{int}(n)g \in \Omega(S')$,*

$$f(ngn^{-1}) = f_N(n)f(g)f_N(n)^{-1}.$$

This is immediate for $n \in T(S)$, by 2.1.3. The two sides define morphisms from $\Omega \cap \text{int}(n)^{-1}\Omega$ to H . It is enough to find an open $V_n \subset \Omega$, containing the unit section, such that $\text{int}(n)V_n \subset \Omega$, and on which $f \circ \text{int}(n)$ and $\text{int}(f_N(n)) \circ f$ agree. By the structure of $\text{Norm}_G(T)$, it suffices to show that if this is true for n' , then it is true for $n = n'w_\alpha$, $\alpha \in \Delta$. Put

$$V_n = \Omega \cap \text{int}(w_\alpha)^{-1}V_{n'}.$$

Then $\text{int}(n)V_n \subset \text{int}(n')V_{n'} \subset \Omega$. If $g \in V_n(S')$, then

$$\text{int}(n)g = \text{int}(n')\text{int}(w_\alpha)g.$$

Now $\text{int}(w_\alpha)g \in V_{n'}(S')$, so the induction hypothesis gives

$$f(\text{int}(n')\text{int}(w_\alpha)g) = \text{int}(f_N(n'))f(\text{int}(w_\alpha)g).$$

Lemma 2.1.7 gives

$$f(\text{int}(w_\alpha)g) = \text{int}(f_N(w_\alpha))f(g),$$

and the conclusion follows.

We now prove Proposition 2.1.2. Let $x, x' \in \Omega(S')$, and write

$$x = vt u, \quad x' = v' t' u',$$

so $xx' = vt(uv')t'u'$. By 2.1.6 and $U^-(S')\Omega(S')U(S') = \Omega(S')$, it suffices to show that $f(uv') = f(u)f(v')$ whenever $uv' \in \Omega(S')$. Choose $n \in \text{Norm}_G(T)(S')$ as in 2.1.5. Then

$$\begin{aligned} f(u) &= f_N(n)^{-1}f(nun^{-1})f_N(n), \\ f(v') &= f_N(n)^{-1}f(nv'n^{-1})f_N(n), \end{aligned}$$

and hence

$$f(u)f(v') = f_N(n)^{-1}f(nun^{-1})f(nv'n^{-1})f_N(n).$$

Since $nun^{-1} \in U^-(S')$ and $nv'n^{-1} \in U(S')$,

$$f(nun^{-1})f(nv'n^{-1}) = f((nun^{-1})(nv'n^{-1})) = f(nuv'n^{-1}).$$

Thus

$$f(u)f(v') = f_N(n)^{-1}f(nuv'n^{-1})f_N(n).$$

If $uv' \in \Omega(S')$, Lemma 2.1.8 finishes the proof.

Remark 2.2.— *Instead of prescribing f_N , one may prescribe a morphism $f_T : T \rightarrow H$ and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$, satisfying the conditions of 1.8.2. Condition (ii) of Theorem 2.1 must then be replaced by:*

(ii') There exists a morphism of groups $F_\alpha : Z_\alpha \rightarrow H$ extending $f_\alpha, f_{-\alpha}, f_T$ and satisfying $F_\alpha(w_\alpha) = h_\alpha$.

We now give the preceding theorem a more explicit form, retaining its notation.

Theorem 2.3.— *Let S be a scheme, let G be a pinned S -group, and let H be an fppf S -sheaf in groups. Suppose given morphisms of groups*

$$f_T : T \longrightarrow H, \quad f_\alpha : U_\alpha \longrightarrow H \quad (\alpha \in \Delta),$$

and sections $h_\alpha \in H(S)$, $\alpha \in \Delta$. There exists a necessarily unique morphism of groups $f : G \rightarrow H$ inducing f_T and the f_α and satisfying $f(w_\alpha) = h_\alpha$ for all $\alpha \in \Delta$ if and only if:

(i) *for every $S' \rightarrow S$, $t \in T(S')$, $\alpha \in \Delta$, and $x \in U_\alpha(S')$,*

$$(1) \quad f_T(t)f_\alpha(x)f_T(t)^{-1} = f_\alpha(\text{int}(t)x);$$

(ii) *for every $\alpha \in \Delta$, $S' \rightarrow S$, and $t \in T(S')$,*

$$(2) \quad h_\alpha f_T(t)h_\alpha^{-1} = f_T(s_\alpha(t)) = f_T(t\alpha^*(\alpha(t)^{-1}));$$

(iii) *for every $(\alpha, \beta) \in \Delta \times \Delta$,*

$$(3) \quad (h_\alpha h_\beta)^{n_{\alpha\beta}} = f_T(t_{\alpha\beta});$$

(iv) for every $\alpha \in \Delta$, recalling that $u_\alpha = \exp_\alpha(X_\alpha)$,

$$(4) \quad (h_\alpha f_\alpha(u_\alpha))^3 = e;$$

(v) for every distinct $\alpha, \beta \in \Delta$, there exists a morphism $f_{\alpha\beta} : U_{\alpha\beta} \rightarrow H$ such that

$$(5) \quad f_{\alpha\beta}|_{U_\alpha} = f_\alpha, \quad f_{\alpha\beta}|_{U_\beta} = f_\beta,$$

and, for every $\gamma \in R_{\alpha\beta}^+$ with $\gamma \neq \alpha$ (respectively $\gamma \neq \beta$) and every $x \in U_\gamma(S')$, $S' \rightarrow S$,

$$(6) \quad \text{int}(h_\alpha)f_{\alpha\beta}(x) = f_{\alpha\beta}(\text{int}(w_\alpha)x)$$

(respectively $\text{int}(h_\beta)f_{\alpha\beta}(x) = f_{\alpha\beta}(\text{int}(w_\beta)x)$).

Proof. Uniqueness follows from 1.9.2. We prove existence.

Lemma 2.3.1.— *There exists a morphism*

$$f_N : \text{Norm}_G(T) \longrightarrow H$$

extending f_T and satisfying $f_N(w_\alpha) = h_\alpha$.

This is precisely 1.8.2, in view of conditions (2) and (3).

Lemma 2.3.2.— *For every $\alpha \in \Delta$, there is a morphism of groups $F_\alpha : Z_\alpha \rightarrow H$ extending f_T and f_α , satisfying $F_\alpha(w_\alpha) = h_\alpha$, and hence extending $f_N|_{\text{Norm}_{Z_\alpha}(T)}$.*

This follows from Exposé XX, 6.2 and conditions (1), (3), and (4).

Lemma 2.3.3.— *Let $(\alpha, \beta) \in \Delta \times \Delta$, $\alpha \neq \beta$. Let $n \in \text{Norm}_{Z_{\alpha\beta}}(T)(S)$ satisfy $n(\alpha) = \alpha$ (respectively $n(\alpha) = \beta$), equivalently $\text{int}(n)U_\alpha = U_\alpha$ (respectively U_β). Then, for every $S' \rightarrow S$ and $x \in U_\alpha(S')$,*

$$\text{int}(f_N(n))f_\alpha(x) = f_\alpha(\text{int}(n)x)$$

(respectively the right-hand side is $f_\beta(\text{int}(n)x)$).

There exist $t \in T(S)$ and a sequence $\{\alpha_i\} \subset \{\alpha, \beta\}$ such that

$$n = tw_{\alpha_k} \cdots w_{\alpha_1}.$$

Condition (1) treats $n = t$, so we may omit t . Proceed by induction on k , and set

$$\gamma_i = s_{\alpha_i} \cdots s_{\alpha_1}(\alpha).$$

If every γ_i is positive, condition (v) yields

$$\text{int}(f_N(w_{\alpha_i} \cdots w_{\alpha_1}))f_\alpha(x) = f_{\alpha\beta}(\text{int}(w_{\alpha_i} \cdots w_{\alpha_1})x),$$

and $i = k$ gives the result. Otherwise there is a $j < k$ such that $\gamma_j \in R^+$ but $\gamma_{j+1} \notin R^+$. Since $\gamma_{j+1} = s_{\alpha_j}(\gamma_j)$, Exposé XXI, 3.3.2 gives $\gamma_j = \alpha_j$, hence γ_j is α or β . Decompose

$$n = n'n'', \quad n' = w_{\alpha_k} \cdots w_{\alpha_{j+1}}, \quad n'' = w_{\alpha_j} \cdots w_{\alpha_1}.$$

Both n' and n'' satisfy the lemma's hypotheses and involve fewer than k reflections, so induction applies.

Lemma 2.3.4.— *Let $\alpha \in R$. Suppose $n, n' \in \text{Norm}_G(T)(S)$ and $\beta, \beta' \in \Delta$ satisfy $n(\alpha) = \beta$ and $n'(\alpha) = \beta'$. Then, for every $x \in U_\alpha(S')$, $S' \rightarrow S$,*

$$\text{int}(f_N(n)^{-1})f_\beta(nxn^{-1}) = \text{int}(f_N(n')^{-1})f_{\beta'}(n'xn'^{-1}).$$

It suffices to prove that if $n(\alpha) = \beta$, with $\alpha, \beta \in \Delta$, then

$$\text{int}(f_N(n)) \circ f_\alpha = f_\beta \circ \text{int}(n).$$

By Tits's lemma (Exposé XXI, 5.6), there are simple roots

$$\alpha_0 = \alpha, \alpha_1, \dots, \alpha_m = \beta$$

and elements $w_i \in W$, $0 \leq i < m$, such that

$$n = w_{m-1} \cdots w_0, \quad w_i(\alpha_i) = \alpha_{i+1},$$

and for each i there is a simple root β_i such that

$$w_i \in W_{\alpha_i \beta_i}, \quad \alpha_{i+1} = \alpha_i \text{ or } \beta_i.$$

Induction reduces the assertion to Lemma 2.3.3.

Lemma 2.3.5.— *There is a family of morphisms of groups $f'_\gamma : U_\gamma \rightarrow H$, $\gamma \in R$, such that:*

- (i) *for $\alpha \in \Delta$, $f'_\alpha = f_\alpha$ and $f'_{-\alpha} = F_\alpha|_{U_{-\alpha}}$;*
- (ii) *if $\alpha, \beta \in \Delta$ and $\gamma \in R_{\alpha\beta}^+$, then $f'_\gamma = f_{\alpha\beta}|_{U_\gamma}$;*
- (iii) *if $n \in \text{Norm}_G(T)(S)$ and $\alpha, \beta \in R$ satisfy $n(\alpha) = \beta$, then*

$$\text{int}(f_N(n))f'_\alpha(x) = f'_\beta(\text{int}(n)x)$$

for every $x \in U_\alpha(S')$, $S' \rightarrow S$.

For every root α , choose $n \in \text{Norm}_G(T)(S)$ with $n(\alpha) \in \Delta$, and define $f'_\alpha(x)$ by the expression in Lemma 2.3.4. This is independent of n , and f'_α is a morphism of groups. Property (iii) holds by construction. The first part of (i) follows by taking $n = e$, and the second by taking $n = w_\alpha$ and using 2.3.2. If $\gamma \in R_{\alpha\beta}^+$, choose $n \in \text{Norm}_{Z_{\alpha\beta}}(T)(S)$ with $n(\gamma) = \alpha$ or β ; then (iii), together with conditions (5) and (6), proves (ii).

2.3.6.— We finish the proof by verifying the conditions of Theorem 2.1 for f_N and f'_α , $\alpha \in R$:

- 2.1(i) follows from 2.3.5(iii);
- 2.1(ii) follows from 2.3.5(i) and 2.3.2;
- 2.1(iii) follows from 2.3.5(ii) and the fact that $f_{\alpha\beta}$ is a morphism of groups.

An immediate corollary is:

Corollary 2.4.— *Let S be a scheme, let G be a pinned S -group of semisimple rank at least 1, and let H be an fppf S -sheaf in groups. For each $(\alpha, \beta) \in \Delta \times \Delta$, let*

$$F_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow H$$

be a morphism of groups. There exists a morphism $f : G \rightarrow H$ inducing the $F_{\alpha\beta}$ if and only if, for every $(\alpha, \beta) \in \Delta \times \Delta$,

$$F_{\alpha\beta}|_{Z_\alpha} = F_{\alpha\alpha}.$$

Necessity is clear. Conversely, put

$$f_T = F_{\alpha\alpha}|_T;$$

this is independent of α , since

$$F_{\alpha\alpha}|_T = F_{\alpha\beta}|_T = F_{\beta\beta}|_T.$$

Set

$$f_\alpha = F_{\alpha\alpha}|_{U_\alpha}, \quad h_\alpha = F_{\alpha\alpha}(w_\alpha), \quad f_{\alpha\beta} = F_{\alpha\beta}|_{U_{\alpha\beta}}.$$

The conditions of Theorem 2.3 hold: (1), (2), and (4) hold in Z_α , while (3), (5), and (6) hold in $Z_{\alpha\beta}$. Thus there is a morphism $f : G \rightarrow H$ extending f_T and all f_α , and satisfying $f(w_\alpha) = h_\alpha$. It agrees with $F_{\alpha\beta}$ on the generators of $Z_{\alpha\beta}$; hence $f|_{Z_{\alpha\beta}} = F_{\alpha\beta}$.

Corollary 2.5.— *Let S be a scheme; let G and G' be two pinned S -groups of semisimple rank 2; let $q > 0$ be an integer such that $x \mapsto x^q$ is an endomorphism of $\mathbf{G}_{a,S}$; and let*

$$h : R(G') \longrightarrow R(G)$$

be a q -morphism of pinned root data. For each $\alpha \in R^+$, choose

$$X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times, \quad X'_{d(\alpha)} \in \Gamma(S, \mathfrak{g}'^{d(\alpha)})^\times,$$

extending the canonical choices for $\alpha \in \Delta$. Suppose:

(i) *if $\Delta = \{\alpha, \beta\}$, then*

$$D_S(h)(t_{\alpha\beta}) = t'_{d(\alpha)d(\beta)};$$

(ii) *for every $\alpha \in \Delta$ and every $\beta \in R^+$, $\beta \neq \alpha$, if $z \in \mathbf{G}_m(S)$ is defined by*

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{s_\alpha(\beta)},$$

then

$$\mathrm{Ad}(w'_{d(\alpha)})X'_{d(\beta)} = z^{q(\beta)}X'_{d(s_\alpha(\beta))};$$

(iii) *there is a morphism of groups $f : U \rightarrow U'$ such that, for every $\alpha \in R^+$, every $S' \rightarrow S$, and every $x \in \mathbf{G}_a(S')$,*

$$f(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)}).$$

Then there is a morphism of pinned groups $g : G \rightarrow G'$ such that $R(g) = h$.

Indeed, define $f_\alpha : U_\alpha \rightarrow G'$ by

$$f_\alpha(\exp(xX_\alpha)) = \exp(x^{q(\alpha)}X'_{d(\alpha)}),$$

and put

$$f_T = D_S(h), \quad h_\alpha = w'_{d(\alpha)}.$$

The conditions of Theorem 2.3 hold: note that

$$q(s_\alpha(\beta)) = q(\beta)$$

by Exposé XXI, 6.8.4, and that

$$D_S(h)(t_\alpha) = t'_{d(\alpha)}$$

always holds. The assertion follows at once.

Remark 2.6.— *Condition (v) of Theorem 2.3 may be made more explicit as follows. One must first verify:*

- (a) *for every word in w_α, w_β whose corresponding word carries α or β to α or β , the corresponding relation of the type occurring in 2.3.3 holds.*

The proof of 2.3.3 shows that it suffices to check the words that are minimal, in the sense that no nontrivial initial subword satisfies the imposed conditions.

If (a) holds, define $f_\gamma : U_\gamma \rightarrow H$ for every $\gamma \in R_{\alpha\beta}^+$ as in 2.3.5. One must then require:

- (b) *the map defined by the f_γ ,*

$$U_{\alpha\beta} = \prod_{\gamma \in R_{\alpha\beta}^+} U_\gamma \longrightarrow H,$$

is a morphism of groups.

By Exposé XXII, 5.5.8, condition (b) follows from:

- (b') *the preceding map respects the commutation relations between U_γ and U_δ , for $\gamma, \delta \in R_{\alpha\beta}^+$, $\gamma > \delta$; namely, the relations involving $C_{i,j,\gamma,\delta}$ in Exposé XXII, 5.5.2.*

Conversely, the conjunction of (a) and (b') is equivalent to condition (v).

These conditions may even be reduced to conditions involving only

$$h_\alpha, \quad h_\beta, \quad f_\alpha(u_\alpha), \quad f_\beta(u_\beta)$$

in H . For example, if $s_\alpha s_\beta s_\alpha(\beta) = \alpha$, a condition of type (a) reads

$$(1) \quad \text{int}(h_\alpha h_\beta h_\alpha) f_\beta(x) = f_\alpha(\text{int}(w_\alpha w_\beta w_\alpha)x).$$

Here $x \in U_\beta(S')$, $S' \rightarrow S$. In particular, for $x = u_\beta$,

$$\text{int}(w_\alpha w_\beta w_\alpha) u_\beta = u_\alpha^z$$

for a certain $z \in \mathbf{G}_m(S)$, and

$$(1') \quad \text{int}(h_\alpha h_\beta h_\alpha) f_\beta(u_\beta) = f_\alpha(u_\alpha)^z.$$

Conversely, taking conditions (i)–(iv) of 2.3 into account, (1') implies (1). If $t \in T(S')$, conjugating (1') by t , and using (i) and (ii), gives (1) for

$$x = \text{int}(t)u_\beta = u_\beta^{\beta(t)}.$$

The morphism $\beta : T \rightarrow \mathbf{G}_{m,S}$ is faithfully flat, so (1) holds for $x \in U_\beta(S')^\times$. It is additive in x , and every section of U_β is locally the sum of two invertible sections. Thus

$$(1') + (i) + (ii) \implies (1).$$

The same argument applies to conditions of type (b). One uses the fact that, for two distinct positive roots γ, γ' , hence linearly independent over $\mathbf{Q}$, the morphism

$$T \longrightarrow \mathbf{G}_{m,S}^2$$

with components γ, γ' is faithfully flat. We leave the details of this transposition to the reader.

3. Groups of Semisimple Rank 2

3.1. Generalities

Lemma 3.1.1.— *Let S be a scheme, let G be a pinned S -group, and let α, β be two roots of G such that $\alpha + \beta \neq 0$.*

(i) *If $\alpha + \beta \notin R$, then*

$$\exp(X_\alpha) \exp(X_\beta) = \exp(X_\beta) \exp(X_\alpha)$$

for every $S' \rightarrow S$, $X_\alpha \in W(\mathfrak{g}^\alpha)(S')$, and $X_\beta \in W(\mathfrak{g}^\beta)(S')$.

(ii) *If neither $\alpha + \beta$ nor $\alpha - \beta$ is a root, then*

$$w_\alpha(z_\alpha) \exp(X_\beta) w_\alpha(z_\alpha)^{-1} = \exp(X_\beta)$$

for every $X_\beta \in W(\mathfrak{g}^\beta)(S')$ and $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, and

$$w_\alpha(z_\alpha) w_\beta(z_\beta) = w_\beta(z_\beta) w_\alpha(z_\alpha)$$

for every $z_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$ and $z_\beta \in W(\mathfrak{g}^\beta)^\times(S')$.

(iii) *Let $X_\alpha \in W(\mathfrak{g}^\alpha)^\times(S')$, $X_\beta \in W(\mathfrak{g}^\beta)^\times(S')$, and $w \in \text{Norm}_G(T)(S')$ satisfy $w(\alpha) = \beta$. Define $z \in \mathbf{G}_m(S')$ by*

$$\text{Ad}(w)X_\alpha = zX_\beta.$$

Then

$$\begin{aligned} \text{int}(w) \exp(xX_\alpha) &= \exp(xzX_\beta), \\ \text{int}(w) \exp(yX_\alpha^{-1}) &= \exp(yz^{-1}X_\beta^{-1}), \\ \text{int}(w)w_\alpha(X_\alpha) &= \beta^*(z)w_\beta(X_\beta). \end{aligned}$$

In particular, if $z = \pm 1$, then

$$\text{int}(w)w_\alpha(X_\alpha) = w_\beta(X_\beta)^z.$$

(iv) *If $t_\alpha = \alpha^*(-1)$ and $t_\beta = \beta^*(-1)$, then*

$$s_\alpha(t_\beta) = t_\beta t_\alpha^{(\beta^*, \alpha)}, \quad \beta(t_\alpha) = (-1)^{(\alpha^*, \beta)}.$$

Proof. Part (i) follows immediately from Exposé XXII, 5.5.2. Part (ii) follows from Exposé XX, 3.1 and from (i) applied to

$$(\beta, \alpha), \quad (\beta, -\alpha), \quad (-\beta, \alpha), \quad (-\beta, -\alpha).$$

Part (iii) is immediate from the definitions; for its last assertion, observe that

$$\beta^*(-1) = w_\beta(X_\beta)^{-2}.$$

Finally, (iv) is tautological.

Proposition 3.1.2 (Groups of type $A_1 \times A_1$).— *Let S be a scheme and G a pinned S -group of type $A_1 \times A_1$. Write $\Delta = R^+ = \{\alpha, \beta\}$.*

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^2 = t_\alpha t_\beta = (w_\beta w_\alpha)^2 = t_{\beta\alpha}.$$

(ii)

$$\mathrm{Ad}(w_\alpha)X_\beta = X_\beta, \quad \mathrm{Ad}(w_\beta)X_\alpha = X_\alpha.$$

(iii) The groups U_α and U_β commute; in other words, U is commutative.*Proof.* Lemma 3.1.1(ii) gives $w_\alpha w_\beta = w_\beta w_\alpha$, whence

$$(w_\alpha w_\beta)^2 = w_\alpha^2 w_\beta^2 = t_\alpha t_\beta,$$

which proves (i). The same lemma gives (ii), and (iii) is Lemma 3.1.1(i).

3.1.3.— Condition (v) of Theorem 2.3 takes the following explicit form. Put

$$v_\alpha = f_\alpha(u_\alpha) \quad (\alpha \in \Delta).$$

Then the two groups of conditions are

$$(A) \quad \begin{cases} h_\alpha v_\beta h_\alpha^{-1} = v_\beta, \\ h_\beta v_\alpha h_\beta^{-1} = v_\alpha, \end{cases} \quad (B) \quad v_\alpha v_\beta = v_\beta v_\alpha.$$

3.2. Groups of Type A_2 **Proposition 3.2.1.**— Let S be a scheme and G a pinned S -group of type A_2 . Write

$$\Delta = \{\alpha, \beta\}, \quad R^+ = \{\alpha, \beta, \alpha + \beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^3 = e = (w_\beta w_\alpha)^3 = t_{\beta\alpha}.$$

(ii) Put $\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}$. Then

$$\begin{aligned} \mathrm{Ad}(w_\alpha)X_\beta &= -X_{\alpha+\beta}, \\ \mathrm{Ad}(w_\alpha)X_{\alpha+\beta} &= X_\beta, \\ \mathrm{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha. \end{aligned}$$

(iii) Put

$$p_{\alpha+\beta}(x) = \exp(xX_{\alpha+\beta}) = \mathrm{int}(w_\beta)p_\alpha(x).$$

Then

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy).$$

3.2.2.— The proof occupies 3.2.2–3.2.7. First,

$$(\beta^*, \alpha) = (\alpha^*, \beta) = -1,$$

so $\alpha(t_\beta) = \beta(t_\alpha) = -1$. Put

$$\mathrm{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}.$$

Then

$$\mathrm{Ad}(w_\beta)X_{\alpha+\beta} = \alpha(t_\beta)X_\alpha = -X_\alpha.$$

Write

$$\mathrm{Ad}(w_\alpha)X_\beta = zX_{\alpha+\beta}, \quad z \in \mathbf{G}_m(S).$$

It follows that

$$\mathrm{Ad}(w_\alpha)X_{\alpha+\beta} = \beta(t_\alpha)z^{-1}X_\beta = -z^{-1}X_\beta.$$

By Exposé XXII, 5.5.2, there is a unique $A \in \mathbf{G}_a(S)$ such that

$$(+) \quad p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy).$$

Thus (ii) and (iii) reduce to proving $A = -z = 1$.

3.2.3.— Conjugating (+) by w_β gives

$$(++) \quad p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy).$$

3.2.4.— By definition,

$$w_\beta p_\alpha(x)w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Expanding w_β , this is

$$\begin{aligned} p_\beta(1)p_{-\beta}(-1)p_\beta(1)p_\alpha(x)p_\beta(-1)p_{-\beta}(1)p_\beta(-1) \\ = p_{\alpha+\beta}(x). \end{aligned}$$

Since p_β commutes with $p_{\alpha+\beta}$, this may be written

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (+) on the left and (++) on the right gives

$$\begin{aligned} p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_\beta(-1) &= p_{\alpha+\beta}(x)p_{-\beta}(1) \\ &\quad \cdot p_\alpha(Ax)p_{-\beta}(-1). \end{aligned}$$

Because $\alpha + 2\beta$, $\alpha - \beta$, and $2\alpha + \beta$ are not roots, this reduces to

$$p_\alpha(x)p_{\alpha+\beta}(Ax) = p_\alpha(Ax)p_{\alpha+\beta}(x).$$

Exposé XXII, 4.1.3 now gives $A = 1$.

3.2.5.— Conjugating (+) by w_α , and using $A = 1$, gives

$$(+++) \quad p_{\alpha+\beta}(zy)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(zy)p_\beta(-z^{-1}xy).$$

3.2.6.— As before,

$$w_\alpha p_\beta(y)w_\alpha^{-1} = p_{\alpha+\beta}(zy).$$

Since p_α and $p_{\alpha+\beta}$ commute,

$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{\alpha+\beta}(zy)p_{-\alpha}(-1).$$

Using (+) and (+++), this becomes

$$p_\beta(y)p_{\alpha+\beta}(-y) = p_{\alpha+\beta}(zy)p_\beta(-z^{-1}y).$$

The two factors commute, and hence $z = -1$.

3.2.7.— It remains to prove (i). One has

$$w_\alpha w_\beta w_\alpha = w_\alpha w_\beta w_\alpha^{-1} w_\alpha^2 = w_{\alpha+\beta}^{-1} t_\alpha,$$

and therefore

$$\begin{aligned} w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_\beta w_{\alpha+\beta}^{-1} t_\alpha w_\beta \\ &= w_\beta w_{\alpha+\beta}^{-1} w_\beta^{-1} s_\beta(t_\alpha) t_\beta \\ &= w_\alpha t_\alpha t_\beta t_\beta = w_\alpha t_\alpha = w_\alpha^{-1}. \end{aligned}$$

Thus

$$(w_\alpha w_\beta)^3 = (w_\beta w_\alpha)^3 = e.$$

Remark 3.2.8.— Put $v_\alpha = f_\alpha(u_\alpha)$. Condition (v) of 2.3 becomes

$$(A) \quad \begin{cases} \text{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \text{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta (h_\beta v_\alpha h_\beta^{-1}), \\ v_\beta (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\beta, \\ v_\alpha (h_\beta v_\alpha h_\beta^{-1}) = (h_\beta v_\alpha h_\beta^{-1}) v_\alpha. \end{cases}$$

Writing $v_{\alpha+\beta} = \text{int}(h_\beta) v_\alpha$, the latter conditions are equivalently

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta}, \\ v_\alpha v_{\alpha+\beta} = v_{\alpha+\beta} v_\alpha, \\ v_\beta v_{\alpha+\beta} = v_{\alpha+\beta} v_\beta. \end{cases}$$

3.3. Groups of Type B_2

Proposition 3.3.1.— Let S be a scheme and G a pinned S -group of type B_2 . Write

$$\Delta = \{\alpha, \beta\}, \quad R^+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^4 = t_\alpha = (w_\beta w_\alpha)^4 = t_{\beta\alpha}.$$

(ii) If

$$\text{Ad}(w_\beta) X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha) X_\beta = X_{2\alpha+\beta},$$

then

$$\begin{aligned} \text{Ad}(w_\alpha) X_{\alpha+\beta} &= -X_{\alpha+\beta}, & \text{Ad}(w_\alpha) X_{2\alpha+\beta} &= X_\beta, \\ \text{Ad}(w_\beta) X_{\alpha+\beta} &= -X_\alpha, & \text{Ad}(w_\beta) X_{2\alpha+\beta} &= X_{2\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(x X_{\alpha+\beta}) = \text{int}(w_\beta) p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(x X_{2\alpha+\beta}) = \text{int}(w_\alpha) p_\beta(x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y) p_\alpha(x) &= p_\alpha(x) p_\beta(y) p_{\alpha+\beta}(xy) p_{2\alpha+\beta}(x^2 y), \\ p_{\alpha+\beta}(y) p_\alpha(x) &= p_\alpha(x) p_{\alpha+\beta}(y) p_{2\alpha+\beta}(2xy). \end{aligned}$$

3.3.2.— The proof occupies 3.3.2–3.3.6. We have

$$(\beta^*, \alpha) = -1, \quad (\alpha^*, \beta) = -2,$$

so $\alpha(t_\beta) = -1$ and $\beta(t_\alpha) = 1$. In particular, $\beta(t_\alpha) = \alpha(t_\alpha) = 1$, so t_α is a section of $\text{Centr}(G)$. Put

$$\text{Ad}(w_\beta)X_\alpha = X_{\alpha+\beta}, \quad \text{Ad}(w_\alpha)X_\beta = X_{2\alpha+\beta}.$$

Then

$$\begin{aligned} \text{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \text{Ad}(w_\alpha)X_{2\alpha+\beta} &= \beta(t_\alpha)X_\beta = X_\beta, \\ \text{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}. \end{aligned}$$

For some $k \in \mathbf{G}_m(S)$,

$$\text{Ad}(w_\alpha)X_{\alpha+\beta} = kX_{\alpha+\beta}.$$

By Exposé XXII, 5.5.2, there are $A, B, C \in \mathbf{G}_a(S)$ such that

$$\begin{aligned} (1) \quad & p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y), \\ (2) \quad & p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Cxy). \end{aligned}$$

It remains to prove

$$A = B = 1, \quad C = 2, \quad k = -1.$$

3.3.3.— Conjugating (1) by w_α gives

$$\begin{aligned} (3) \quad & p_{2\alpha+\beta}(y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{2\alpha+\beta}(y) \\ & \cdot p_{\alpha+\beta}(Aky)p_\beta(Bx^2y). \end{aligned}$$

Transforming (2) in the same way gives

$$(4) \quad p_{\alpha+\beta}(ky)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{\alpha+\beta}(ky)p_\beta(Cxy).$$

Conjugating (1) by w_β gives

$$\begin{aligned} (5) \quad & p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y) \\ & \cdot p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y). \end{aligned}$$

3.3.4.— Write

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x).$$

Because $\alpha + 2\beta$ is not a root,

$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) on the left and (5) on the right yields

$$\begin{aligned} & p_\alpha(x)p_\beta(1)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_\beta(-1) \\ & = p_{\alpha+\beta}(x)p_{-\beta}(1)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{-\beta}(-1). \end{aligned}$$

After commuting the indicated factors and applying (2), this becomes

$$\begin{aligned} & p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2) = p_\alpha(Ax)p_{\alpha+\beta}(x) \\ & \cdot p_{2\alpha+\beta}((AC - B)x^2). \end{aligned}$$

Hence

$$A = 1, \quad C = 2B.$$

3.3.5.— Now write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{2\alpha+\beta}(y).$$

Since $3\alpha + \beta$ is not a root,

$$p_\alpha(1) p_\beta(y) p_\alpha(-1) = p_{-\alpha}(1) p_{2\alpha+\beta}(y) p_{-\alpha}(-1).$$

Using (1) on the left and (3) on the right gives

$$p_\beta(y) p_{\alpha+\beta}(-Ay) p_{2\alpha+\beta}(By) = p_{2\alpha+\beta}(y) p_{\alpha+\beta}(Aky) p_\beta(By).$$

The three groups commute, so

$$B = 1, \quad -A = Ak.$$

Consequently

$$A = B = 1, \quad C = 2, \quad k = -1.$$

3.3.6.— We have proved (ii) and (iii). Since

$$s_\beta(t_\alpha) = t_\alpha t_\beta^{(\alpha^*, \beta)} = t_\alpha,$$

one obtains successively

$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_{2\alpha+\beta} t_\alpha, \\ w_\beta w_\alpha w_\beta w_\alpha w_\beta &= w_{2\alpha+\beta} t_\alpha t_\beta, \\ w_\alpha w_\beta w_\alpha w_\beta w_\alpha w_\beta w_\alpha &= w_\beta^{-1} t_\alpha. \end{aligned}$$

⁹ It follows that

$$(w_\beta w_\alpha)^4 = t_\alpha, \quad (w_\alpha w_\beta)^4 = s_\beta(t_\alpha) = t_\alpha.$$

Remark 3.3.7.— Put

$$v_{\alpha+\beta} = \text{int}(h_\beta) v_\alpha, \quad v_{2\alpha+\beta} = \text{int}(h_\alpha) v_\beta.$$

Condition (v) of 2.3 becomes

$$(A) \quad \begin{cases} \text{int}(h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \text{int}(h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and

$$(B) \quad \begin{cases} v_\beta v_\alpha = v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta}, \\ v_{\alpha+\beta} v_\alpha = v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2, \\ v_{\alpha+\beta} v_\beta = v_\beta v_{\alpha+\beta}, \\ v_{2\alpha+\beta} v_\alpha = v_\alpha v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_\beta = v_\beta v_{2\alpha+\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} = v_{\alpha+\beta} v_{2\alpha+\beta}. \end{cases}$$

⁹ *Editors' note.* The French prints $(\alpha^*, \beta) = 2$; the convention used immediately above gives -2 . The conclusion is unchanged because $t_\beta^2 = e$.

3.4. Groups of Type G_2

Proposition 3.4.1.— *Let S be a scheme and G a pinned S -group of type G_2 . Write*

$$\Delta = \{\alpha, \beta\}, \quad R^+ = \{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}.$$

(i)

$$t_{\alpha\beta} = (w_\alpha w_\beta)^6 = e = (w_\beta w_\alpha)^6 = t_{\beta\alpha}.$$

(ii) If

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, \end{aligned}$$

then

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= X_\beta, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} &= X_{3\alpha+2\beta}, & \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= -X_\alpha, \\ \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}. \end{aligned}$$

(iii) Put

$$\begin{aligned} p_{\alpha+\beta}(x) &= \exp(xX_{\alpha+\beta}) = \operatorname{int}(w_\beta)p_\alpha(x), \\ p_{2\alpha+\beta}(x) &= \exp(xX_{2\alpha+\beta}) = \operatorname{int}(w_\alpha w_\beta)p_\alpha(x), \\ p_{3\alpha+\beta}(x) &= \exp(xX_{3\alpha+\beta}) = \operatorname{int}(w_\alpha)p_\beta(-x), \\ p_{3\alpha+2\beta}(x) &= \exp(xX_{3\alpha+2\beta}) = \operatorname{int}(w_\beta w_\alpha)p_\beta(-x). \end{aligned}$$

Then

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(2xy)p_{3\alpha+\beta}(3x^2y)p_{3\alpha+2\beta}(3xy^2), \\ p_{2\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(3xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy), \\ p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) &= p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(3xy). \end{aligned}$$

3.4.2.— The proof occupies 3.4.2–3.4.9. Since

$$(\beta^*, \alpha) = -1, \quad (\alpha^*, \beta) = -3,$$

we have $\beta(t_\alpha) = \alpha(t_\beta) = -1$. Define the four nonsimple root vectors as in (ii). Then

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= \alpha(t_\beta)X_\alpha = -X_\alpha, \\ \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= \alpha(t_\alpha)\beta(t_\alpha)X_{\alpha+\beta} = -X_{\alpha+\beta}, \\ \operatorname{Ad}(w_\alpha)X_{3\alpha+\beta} &= -\beta(t_\alpha)X_\beta = X_\beta, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= \alpha(t_\beta)^3\beta(t_\beta)X_{3\alpha+\beta} = -X_{3\alpha+\beta}. \end{aligned}$$

Since $(3\alpha + 2\beta) \pm \alpha$ and $(2\alpha + \beta) \pm \beta$ are not roots,

$$\operatorname{Ad}(w_\alpha)X_{3\alpha+2\beta} = X_{3\alpha+2\beta}, \quad \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} = X_{2\alpha+\beta}.$$

3.4.3.— By Exposé XXII, 5.5.2, there are sections

$$A, B, C, D, E, F, G, H, J \in \mathbf{G}_a(S)$$

such that

- (1) $p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(Axy)p_{2\alpha+\beta}(Bx^2y)p_{3\alpha+\beta}(Cx^3y)p_{3\alpha+2\beta}(Dx^3y^2),$
- (2) $p_{\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(Exy)p_{3\alpha+\beta}(Fx^2y)p_{3\alpha+2\beta}(Gxy^2),$
- (3) $p_{2\alpha+\beta}(y)p_\alpha(x) = p_\alpha(x)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hxy),$
- (4) $p_{3\alpha+\beta}(y)p_\beta(x) = p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(Jxy).$

3.4.4.— Conjugating (1), (3), and (4) by w_β gives

- (5)
$$p_{-\beta}(-y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{-\beta}(-y)p_\alpha(-Axy)p_{2\alpha+\beta}(Bx^2y) \\ \cdot p_{3\alpha+2\beta}(Cx^3y)p_{3\alpha+\beta}(-Dx^3y^2),$$
- (6)
$$p_{2\alpha+\beta}(y)p_{\alpha+\beta}(x) = p_{\alpha+\beta}(x)p_{2\alpha+\beta}(y)p_{3\alpha+2\beta}(Hxy),$$
- (7)
$$p_{3\alpha+2\beta}(y)p_{-\beta}(-x) = p_{-\beta}(-x)p_{3\alpha+2\beta}(y)p_{3\alpha+\beta}(-Jxy).$$

Conjugating (1) by w_α gives

- (8)
$$p_{3\alpha+\beta}(-y)p_{-\alpha}(-x) = p_{-\alpha}(-x)p_{3\alpha+\beta}(-y)p_{2\alpha+\beta}(Axy)p_{\alpha+\beta}(-Bx^2y) \\ \cdot p_\beta(Cx^3y)p_{3\alpha+\beta}(Dx^3y^2).$$

3.4.5.— The identity

$$w_\beta p_\alpha(x) w_\beta^{-1} = p_{\alpha+\beta}(x)$$

may be written, since $\alpha + 2\beta$ is not a root,

- (9)
$$p_\beta(1)p_\alpha(x)p_\beta(-1) = p_{-\beta}(1)p_{\alpha+\beta}(x)p_{-\beta}(-1).$$

Using (1) and then (4), its left side becomes

- (10)
$$p_\alpha(x)p_{\alpha+\beta}(Ax)p_{2\alpha+\beta}(Bx^2)p_{3\alpha+\beta}(Cx^3)p_{3\alpha+2\beta}((D - CJ)x^3).$$

Using (5) and then (7), its right side becomes

$$p_{\alpha+\beta}(x)p_\alpha(Ax)p_{2\alpha+\beta}(-Bx^2)p_{3\alpha+2\beta}(-Cx^3)p_{3\alpha+\beta}((CJ - D)x^3).$$

Applying (2) transforms this into

- (12)
$$p_\alpha(Ax)p_{\alpha+\beta}(x)p_{2\alpha+\beta}((AE - B)x^2)p_{3\alpha+\beta}((A^2F + CJ - D)x^3) \\ \cdot p_{3\alpha+2\beta}((AG - C)x^3).$$

Comparing (10) and (12) gives

$$A = 1, \quad E = 2B, \quad C + D = F + CJ, \quad F = G.$$

3.4.6.— Write

$$w_\alpha p_\beta(y) w_\alpha^{-1} = p_{3\alpha+\beta}(-y).$$

Since $4\alpha + \beta$ is not a root,

- (13)
$$p_\alpha(1)p_\beta(y)p_\alpha(-1) = p_{-\alpha}(1)p_{3\alpha+\beta}(-y)p_{-\alpha}(-1).$$

Using (1), the left side is

$$p_\beta(y)p_{\alpha+\beta}(-Ay)p_{2\alpha+\beta}(By)p_{3\alpha+\beta}(-Cy)p_{3\alpha+2\beta}(-Dy^2).$$

Using (8), (6), and (4), the right side is

$$p_\beta(Cy)p_{\alpha+\beta}(-By)p_{2\alpha+\beta}(Ay)p_{3\alpha+\beta}(-y)p_{3\alpha+2\beta}((D-CJ-ABH)y^2).$$

Thus

$$C = 1, \quad A = B, \quad D - CJ - ABH = -D.$$

Together with the preceding identities, this yields

$$A = B = C = 1, \quad E = 2, \quad F = G, \quad D + 1 = F + J, \quad 2D = H + J.$$

3.4.7.— From

$$w_\beta p_{3\alpha+\beta}(x)w_\beta^{-1} = p_{3\alpha+2\beta}(x)$$

one obtains

$$p_{3\alpha+\beta}(x)p_{3\alpha+2\beta}(-Jx) = p_{3\alpha+2\beta}(x)p_{3\alpha+\beta}(-Jx),$$

and therefore $J = -1$.

3.4.8.— Finally,

$$w_\alpha p_{\alpha+\beta}(y)w_\alpha^{-1} = p_{2\alpha+\beta}(y),$$

or equivalently

$$p_\alpha(1)p_{\alpha+\beta}(y)p_\alpha(-1) = p_{-\alpha}(1)p_\alpha(-1)p_{2\alpha+\beta}(y)p_\alpha(1)p_{-\alpha}(-1).$$

Using (2) on the left and (3) on the right gives

$$\begin{aligned} p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-Ey)p_{3\alpha+\beta}(Fy)p_{3\alpha+2\beta}(-Gy^2) \\ = p_{-\alpha}(1)p_{2\alpha+\beta}(y)p_{3\alpha+\beta}(Hy)p_{-\alpha}(-1). \end{aligned}$$

Commuting $p_{-\alpha}(-1)$ first past $p_{3\alpha+\beta}(Hy)$, and then past $p_{2\alpha+\beta}(y)$, introduces no new $p_{3\alpha+\beta}$ -factor. Comparing those factors gives $F = H$. Hence

$$2D = D + 1, \quad D = 1, \quad F = G = H = 2D - J = 3.$$

This determines all the coefficients and proves (iii).

3.4.9.— For (i), one obtains successively

$$\begin{aligned} w_\alpha w_\beta w_\alpha &= w_{3\alpha+\beta}^{-1} t_\alpha, \\ w_\beta (w_\alpha w_\beta)^2 &= w_{3\alpha+2\beta}^{-1} t_\alpha, \\ w_\alpha (w_\beta w_\alpha)^3 &= w_{3\alpha+2\beta}^{-1}, \\ w_\beta (w_\alpha w_\beta)^4 &= w_{3\alpha+\beta} t_\beta, \\ w_\alpha (w_\beta w_\alpha)^5 &= w_\beta^{-1}. \end{aligned}$$

Consequently

$$(w_\alpha w_\beta)^6 = (w_\beta w_\alpha)^6 = e.$$

Remark 3.4.10.— Condition (v) of 2.3 consists first of

$$(A) \quad \begin{cases} \text{int}(h_\alpha h_\beta h_\alpha h_\beta h_\alpha) v_\beta = v_\beta, \\ \text{int}(h_\beta h_\alpha h_\beta h_\alpha h_\beta) v_\alpha = v_\alpha, \end{cases}$$

and, after setting

$$\begin{aligned} v_{\alpha+\beta} &= \text{int}(h_\beta)v_\alpha, & v_{2\alpha+\beta} &= \text{int}(h_\alpha h_\beta)v_\alpha, \\ v_{3\alpha+\beta} &= \text{int}(h_\alpha)v_\beta^{-1}, & v_{3\alpha+2\beta} &= \text{int}(h_\beta h_\alpha)v_\beta^{-1}, \end{aligned}$$

of the commutation relations

$$\begin{aligned} v_\beta v_\alpha &= v_\alpha v_\beta v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+\beta} v_{3\alpha+2\beta}, \\ v_{\alpha+\beta} v_\alpha &= v_\alpha v_{\alpha+\beta} v_{2\alpha+\beta}^2 v_{3\alpha+\beta}^3 v_{3\alpha+2\beta}^3, \\ v_{2\alpha+\beta} v_\alpha &= v_\alpha v_{2\alpha+\beta} v_{3\alpha+\beta}^3, \\ v_{3\alpha+\beta} v_\alpha &= v_\alpha v_{3\alpha+\beta}, & v_{3\alpha+2\beta} v_\alpha &= v_\alpha v_{3\alpha+2\beta}, \\ v_{\alpha+\beta} v_\beta &= v_\beta v_{\alpha+\beta}, & v_{2\alpha+\beta} v_\beta &= v_\beta v_{2\alpha+\beta}, \\ v_{3\alpha+\beta} v_\beta &= v_\beta v_{3\alpha+\beta} v_{3\alpha+2\beta}^{-1}, \\ (B) \quad v_{3\alpha+2\beta} v_\beta &= v_\beta v_{3\alpha+2\beta}, \\ v_{2\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{2\alpha+\beta} v_{3\alpha+2\beta}^3, \\ v_{3\alpha+\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta} v_{\alpha+\beta} &= v_{\alpha+\beta} v_{3\alpha+2\beta}, \\ v_{3\alpha+\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+\beta}, \\ v_{3\alpha+2\beta} v_{2\alpha+\beta} &= v_{2\alpha+\beta} v_{3\alpha+2\beta}, \\ v_{3\alpha+2\beta} v_{3\alpha+\beta} &= v_{3\alpha+\beta} v_{3\alpha+2\beta}. \end{aligned}$$

3.5. Explicit Form of the Generators-and-Relations Theorem

Theorem 3.5.1.— *Let S be a scheme and G a pinned S -group, with maximal torus T , system of simple roots Δ , and elements*

$$u_\alpha \in U_\alpha(S)^\times, \quad w_\alpha \in \text{Norm}_G(T)(S) \quad (\alpha \in \Delta)$$

defined by the pinning. Let H be an fppf S -sheaf in groups, let

$$f_T : T \rightarrow H, \quad f_\alpha : U_\alpha \rightarrow H \quad (\alpha \in \Delta)$$

be morphisms of groups, and let $h_\alpha \in H(S)$. Put

$$v_\alpha = f_\alpha(u_\alpha).$$

There exists a morphism of groups $f : G \rightarrow H$ extending f_T and all the f_α , and satisfying $f(w_\alpha) = h_\alpha$, if and only if:

(i) *for every $S' \rightarrow S$, $\alpha \in \Delta$, $t \in T(S')$, and $x \in U_\alpha(S')$,*

$$(1) \quad \text{int}(f_T(t))f_\alpha(x) = f_\alpha(\text{int}(t)x) = f_\alpha(x^{\alpha(t)});$$

(ii) *for every $\alpha \in \Delta$, every $S' \rightarrow S$, and every $t \in T(S')$,*

$$(2) \quad \text{int}(h_\alpha)f_T(t) = f_T(s_\alpha(t)) = f_T(t\alpha^*(\alpha(t)^{-1}));$$

(iii) for every $\alpha \in \Delta$,

$$(3_1) \quad h_\alpha^2 = f_T(\alpha^*(-1)),$$

and

$$(4) \quad (h_\alpha v_\alpha)^3 = e;$$

(iv) for distinct $\alpha, \beta \in \Delta$:

(a) the appropriate braid relation is

$$(3_2) \quad \begin{cases} (h_\alpha h_\beta)^2 = f_T(\alpha^*(-1)\beta^*(-1)), & (\alpha^*, \beta) = 0, \\ (h_\alpha h_\beta)^3 = e, & (\alpha^*, \beta) = -1, \\ (h_\alpha h_\beta)^4 = f_T(\alpha^*(-1)), & (\alpha^*, \beta) = -2, \\ (h_\alpha h_\beta)^6 = e, & (\alpha^*, \beta) = -3; \end{cases}$$

(b) the corresponding relations (A) and (B) of 3.1.3, 3.2.8, 3.3.7, or 3.4.10 hold.

When it exists, f is necessarily unique.

This follows immediately from Theorem 2.3, Remark 2.6, and the calculations in each rank-two case.

Remark 3.5.2.— The result may equivalently be presented by giving morphisms

$$a_\alpha : T \cdot U_\alpha \longrightarrow H, \quad b_\alpha : \text{Norm}_{Z_\alpha}(T) \longrightarrow H \quad (\alpha \in \Delta),$$

and putting

$$h_\alpha = b_\alpha(w_\alpha), \quad v_\alpha = a_\alpha(u_\alpha).$$

The required conditions are:

- (1) all the a_α and b_α have the same restriction to T ;
- (2) condition (4) and all the conditions in (iv) of Theorem 3.5.1 hold.

3.5.3.— Further applications will be given in the next Exposé. Here is one. Theorem 3.5.1 describes G by generators and relations in the category of fppf S -sheaves. More explicitly, for every $S' \rightarrow S$, let $H(S')$ be the group generated by

$$T(S'), \quad U_\alpha(S') \ (\alpha \in R), \quad w_\alpha \ (\alpha \in R),$$

subject to relations analogous to

$$(1), (2), (3_1), (4), (3_2), (A), (B).$$

Then G is the sheaf associated with the presheaf $S' \mapsto H(S')$.

In particular, if S' is the spectrum of an algebraically closed field k , then $G(S') = H(S')$. This follows immediately from the Nullstellensatz in the form: an fppf-covering sieve over an algebraically closed field is trivial. Thus Theorem 3.5.1 gives an explicit presentation by generators and relations of the abstract group $G(k)$.

4. Uniqueness of Pinned Groups: Fundamental Theorem

Theorem 4.1.— *Let S be a nonempty scheme. The functor R of 1.6 is fully faithful. More precisely, let G, G' be pinned S -groups, let $p > 0$ be an integer such that $x \mapsto x^p$ is an endomorphism of $\mathbf{G}_{a,S}$, and let*

$$h : R(G') \longrightarrow R(G)$$

be a p -morphism of pinned root data. There is a unique morphism of pinned groups $f : G \rightarrow G'$ such that $R(f) = h$.

Uniqueness was proved in 1.9, so only existence remains. By hypothesis there are a bijection $d : R \xrightarrow{\sim} R'$ and a map

$$q : R \longrightarrow \{p^n \mid n \in \mathbf{N}\}$$

such that, for every $\alpha \in R$,

$$h(d(\alpha)) = q(\alpha)\alpha, \quad {}^th(\alpha^*) = q(\alpha)d(\alpha)^*.$$

In particular, G and G' have the same semisimple rank.

4.1.1.— Suppose that both semisimple ranks are 0. Then G, G' are tori:

$$G = T = D_S(M), \quad G' = T' = D_S(M'),$$

and $h : M' \rightarrow M$ is an ordinary homomorphism. Take $f = D_S(h)$.

4.1.2.— Suppose that both semisimple ranks are 1, and put

$$f_T = D_S(h) : T \longrightarrow T'.$$

Writing $\alpha' = d(\alpha)$, the diagram

$$\begin{array}{ccccc} \mathbf{G}_{m,S} & \xrightarrow{\alpha^*} & T & \xrightarrow{\alpha} & \mathbf{G}_{m,S} \\ \downarrow q(\alpha) & & \downarrow f_T & & \downarrow q(\alpha) \\ \mathbf{G}_{m,S} & \xrightarrow{\alpha'^*} & T' & \xrightarrow{\alpha'} & \mathbf{G}_{m,S} \end{array}$$

commutes. Apply Exposé XX, 4.1.

4.1.3.— Suppose that both semisimple ranks are 2. Exposé XXI, 7.5.3 lists every possible h ; we verify the conditions of Corollary 2.5 in each case. Write

$$\Delta = \{\alpha, \beta\}, \quad \Delta' = \{\alpha', \beta'\}, \quad d(\alpha) = \alpha', \quad d(\beta) = \beta'.$$

4.1.4. G and G' of type $A_1 \times A_1$.— Here

$$h(\alpha') = q\alpha, \quad h(\beta') = q_1\beta.$$

Conditions 2.5(ii) and (iii) follow from Proposition 3.1.2(ii),(iii). For (i),

$$\begin{aligned} D_S(h)t_{\alpha\beta} &= D_S(h)(t_\alpha t_\beta) \\ &= {}^th(\alpha^*)(-1) {}^th(\beta^*)(-1) \\ &= \alpha'^*((-1)^q) \beta'^*((-1)^{q_1}). \end{aligned}$$

The assumption on $x \mapsto x^p$, with $S \neq \emptyset$, implies that $p = 1$ or S has characteristic p . Thus $(-1)^q = -1$: if q is even, then $p = 2$ and $1 = -1$. Consequently

$$D_S(h)t_{\alpha\beta} = \alpha'^*(-1)\beta'^*(-1) = t'_{\alpha'\beta'}.$$

4.1.5. G and G' of type A_2 .— Here

$$h(\alpha') = q\alpha, \quad h(\beta') = q\beta.$$

Put

$$X_{\alpha+\beta} = \text{Ad}(w_\beta)X_\alpha, \quad X'_{\alpha'+\beta'} = \text{Ad}(w'_{\beta'})X'_{\alpha'}.$$

Condition (i) follows as above from Proposition 3.2.1(i), and (ii) follows immediately from Proposition 3.2.1(ii). For (iii), the map

$$p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z) \mapsto p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(z^q)$$

must be a homomorphism. The only nontrivial relation is

$$p_\beta(y)p_\alpha(x) = p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy),$$

which maps to

$$p'_{\beta'}(y^q)p'_{\alpha'}(x^q) = p'_{\alpha'}(x^q)p'_{\beta'}(y^q)p'_{\alpha'+\beta'}(x^qy^q).$$

4.1.6.— The same argument treats types B_2 and G_2 when the radical exponents are equal, using Propositions 3.3.1 and 3.4.1. It remains to treat the two exceptional cases in Exposé XXI, 7.5.3.

4.1.7. Type B_2 , characteristic 2, $q(\alpha) = 2q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta\}$$

and

$$\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta'\},$$

where α and β' are the short simple roots. We have

$$\begin{aligned} h(\alpha') &= 2q\alpha, & h(\beta') &= q\beta, \\ h(\alpha' + \beta') &= q(2\alpha + \beta), & h(\alpha' + 2\beta') &= 2q(\alpha + \beta), \end{aligned}$$

so

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 2\beta', & q(\alpha + \beta) &= 2q, \\ d(2\alpha + \beta) &= \alpha' + \beta', & q(2\alpha + \beta) &= q. \end{aligned}$$

Put

$$\begin{aligned} X_{\alpha+\beta} &= \text{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \text{Ad}(w_\alpha)X_\beta, \\ X'_{\alpha'+\beta'} &= \text{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \text{Ad}(w'_{\beta'})X'_{\alpha'}. \end{aligned}$$

For (i), characteristic 2 gives $-1 = 1$, hence

$$t_{\alpha\beta} = t_\alpha = \alpha'^*(-1) = e = \beta'^*(-1) = t'_{\beta'} = t'_{\alpha'\beta'}.$$

For (ii), the two defining identities are

$$\begin{aligned} \text{Ad}(w_\alpha)X_\beta &= X_{2\alpha+\beta}, & \text{Ad}(w'_{\alpha'})X'_{\beta'} &= X'_{\alpha'+\beta'} = X'_{d(2\alpha+\beta)}, \\ \text{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \text{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{\alpha'+2\beta'} = X'_{d(\alpha+\beta)}. \end{aligned}$$

Proposition 3.3.1(ii), together with $-1 = 1$, gives the four remaining identities:

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha+\beta)}, \\ \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= X_\beta, & \operatorname{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= X'_{d(\beta)}, \\ \operatorname{Ad}(w_\beta)X_{\alpha+\beta} &= X_\alpha, & \operatorname{Ad}(w'_{\beta'})X'_{d(\alpha+\beta)} &= X'_{d(\alpha)}, \\ \operatorname{Ad}(w_\beta)X_{2\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{d(2\alpha+\beta)} &= X'_{d(2\alpha+\beta)}. \end{aligned}$$

For (iii), the sole nontrivial relations are

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y), \\ p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(x'y')p'_{\alpha'+2\beta'}(x'y'^2). \end{aligned}$$

The proposed map is

$$\begin{aligned} p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(z)p_{2\alpha+\beta}(t) \\ \longmapsto p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)p'_{\alpha'+2\beta'}(z^{2q})p'_{\alpha'+\beta'}(t^q). \end{aligned}$$

It respects the first relation because the second, with $y' = x^{2q}$ and $x' = y^q$, reads

$$\begin{aligned} p'_{\beta'}(y^q)p'_{\alpha'}(x^{2q}) &= p'_{\alpha'}(x^{2q})p'_{\beta'}(y^q)p'_{\alpha'+2\beta'}((xy)^{2q}) \\ &\quad \cdot p'_{\alpha'+\beta'}((x^2y)^q). \end{aligned}$$

4.1.8. Type G_2 , characteristic 3, $q(\alpha) = 3q$, $q(\beta) = q$.— The positive roots are

$$\{\alpha, \beta, \alpha + \beta, 2\alpha + \beta, 3\alpha + \beta, 3\alpha + 2\beta\}$$

and

$$\{\alpha', \beta', \alpha' + \beta', \alpha' + 2\beta', \alpha' + 3\beta', 2\alpha' + 3\beta'\},$$

where again α and β' are short. We have

$$\begin{aligned} h(\alpha') &= 3q\alpha, & h(\beta') &= q\beta, \\ h(\alpha' + \beta') &= q(3\alpha + \beta), & h(\alpha' + 2\beta') &= q(3\alpha + 2\beta), \\ h(\alpha' + 3\beta') &= 3q(\alpha + \beta), & h(2\alpha' + 3\beta') &= 3q(2\alpha + \beta), \end{aligned}$$

and hence

$$\begin{aligned} d(\alpha + \beta) &= \alpha' + 3\beta', & q(\alpha + \beta) &= 3q, \\ d(2\alpha + \beta) &= 2\alpha' + 3\beta', & q(2\alpha + \beta) &= 3q, \\ d(3\alpha + \beta) &= \alpha' + \beta', & q(3\alpha + \beta) &= q, \\ d(3\alpha + 2\beta) &= \alpha' + 2\beta', & q(3\alpha + 2\beta) &= q. \end{aligned}$$

Define

$$\begin{aligned} X_{\alpha+\beta} &= \operatorname{Ad}(w_\beta)X_\alpha, & X_{2\alpha+\beta} &= \operatorname{Ad}(w_\alpha)X_{\alpha+\beta}, \\ X_{3\alpha+\beta} &= -\operatorname{Ad}(w_\alpha)X_\beta, & X_{3\alpha+2\beta} &= \operatorname{Ad}(w_\beta)X_{3\alpha+\beta}, \\ X'_{\alpha'+\beta'} &= -\operatorname{Ad}(w'_{\alpha'})X'_{\beta'}, & X'_{\alpha'+2\beta'} &= \operatorname{Ad}(w'_{\beta'})X'_{\alpha'+\beta'}, \\ X'_{\alpha'+3\beta'} &= \operatorname{Ad}(w'_{\beta'})X'_{\alpha'}, & X'_{2\alpha'+3\beta'} &= \operatorname{Ad}(w'_{\alpha'})X'_{\alpha'+3\beta'}. \end{aligned}$$

Condition (i) follows from Proposition 3.4.1(i). Four instances of (ii) hold by definition:

$$\begin{aligned} \operatorname{Ad}(w_\beta)X_\alpha &= X_{\alpha+\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{\alpha'} &= X'_{d(\alpha+\beta)}, \\ \operatorname{Ad}(w_\alpha)X_\beta &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{\beta'} &= -X'_{d(3\alpha+\beta)}, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+\beta} &= X_{3\alpha+2\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{d(3\alpha+\beta)} &= X'_{d(3\alpha+2\beta)}, \\ \operatorname{Ad}(w_\alpha)X_{\alpha+\beta} &= X_{2\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{d(\alpha+\beta)} &= X'_{d(2\alpha+\beta)}. \end{aligned}$$

The remaining instances follow from Proposition 3.4.1(ii); for example,

$$\begin{aligned} \operatorname{Ad}(w_\alpha)X_{2\alpha+\beta} &= -X_{\alpha+\beta}, & \operatorname{Ad}(w'_{\alpha'})X'_{d(2\alpha+\beta)} &= -X'_{d(\alpha+\beta)}, \\ \operatorname{Ad}(w_\beta)X_{3\alpha+2\beta} &= -X_{3\alpha+\beta}, & \operatorname{Ad}(w'_{\beta'})X'_{d(3\alpha+2\beta)} &= -X'_{d(3\alpha+\beta)}. \end{aligned}$$

For (iii), in characteristic 3, the nontrivial relations in U are

$$\begin{aligned} p_\beta(y)p_\alpha(x) &= p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(xy)p_{2\alpha+\beta}(x^2y)p_{3\alpha+\beta}(x^3y)p_{3\alpha+2\beta}(x^3y^2), \\ p_{\alpha+\beta}(y)p_\alpha(x) &= p_\alpha(x)p_{\alpha+\beta}(y)p_{2\alpha+\beta}(-xy), \\ p_{3\alpha+\beta}(y)p_\beta(x) &= p_\beta(x)p_{3\alpha+\beta}(y)p_{3\alpha+2\beta}(-xy). \end{aligned}$$

The corresponding relations in U' are

$$\begin{aligned} p'_{\alpha'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'}(y')p'_{\alpha'+\beta'}(-x'y')p'_{\alpha'+2\beta'}(-x'^2y') \\ &\quad \cdot p'_{\alpha'+3\beta'}(-x'^3y')p'_{2\alpha'+3\beta'}(-x'^3y'^2), \\ p'_{\alpha'+\beta'}(y')p'_{\beta'}(x') &= p'_{\beta'}(x')p'_{\alpha'+\beta'}(y')p'_{\alpha'+2\beta'}(x'y'), \\ p'_{\alpha'+3\beta'}(y')p'_{\alpha'}(x') &= p'_{\alpha'}(x')p'_{\alpha'+3\beta'}(y')p'_{2\alpha'+3\beta'}(x'y'). \end{aligned}$$

Define $\phi : U \rightarrow U'$ by

$$\begin{aligned} &\phi(p_\alpha(x)p_\beta(y)p_{\alpha+\beta}(t)p_{2\alpha+\beta}(u)p_{3\alpha+\beta}(v)p_{3\alpha+2\beta}(w)) \\ &= p'_{\alpha'}(x^{3q})p'_{\beta'}(y^q)p'_{\alpha'+3\beta'}(t^{3q})p'_{2\alpha'+3\beta'}(u^{3q})p'_{\alpha'+\beta'}(v^q)p'_{\alpha'+2\beta'}(w^q). \end{aligned}$$

Substituting x^{3q}, y^q in the three displayed relations for U' yields exactly the images of the three relations for U ; thus ϕ is a homomorphism.

4.1.9. Semisimple rank greater than 2.— For $\alpha \in \Delta$, put $\alpha' = d(\alpha)$. For every (α, β) , consider the pinned rank-at-most-two groups

$$Z_{\alpha\beta}, \quad Z'_{\alpha'\beta'}.$$

The homomorphism $M' \rightarrow M$ underlying h induces a p -morphism

$$h_{\alpha\beta} : R(Z'_{\alpha'\beta'}) \longrightarrow R(Z_{\alpha\beta}).$$

By the preceding cases, there is a morphism of pinned groups

$$f_{\alpha\beta} : Z_{\alpha\beta} \longrightarrow Z'_{\alpha'\beta'}$$

with $R(f_{\alpha\beta}) = h_{\alpha\beta}$. The restrictions

$$f_{\alpha\beta}|_{Z_\alpha}, \quad f_{\alpha\beta} : Z_\alpha \rightarrow Z'_{\alpha'}$$

correspond to the same morphism of pinned root data, and therefore coincide by uniqueness. Corollary 2.4 now glues the $f_{\alpha\beta}$ to a morphism $f : G \rightarrow G'$. It is a morphism of pinned groups and satisfies $R(f) = h$, completing the proof.

5. Corollaries of the Fundamental Theorem

The most important is the following.

Corollary 5.1.— *Let S be a nonempty scheme, let G, G' be pinned S -groups, and let*

$$h : R(G') \xrightarrow{\sim} R(G)$$

be an isomorphism of pinned root data. There is a unique isomorphism of pinned S -groups $f : G \xrightarrow{\sim} G'$ such that $R(f) = h$.

This also follows from Theorem 3.5.1, because its relations can be written using only $R(G)$. It also follows from the elementary part of the proof of Theorem 4.1, without the exceptional isogenies of 4.1.7 and 4.1.8.

Corollary 5.2 (Uniqueness theorem).— *Let S be a scheme and let G, G' be two splittable S -groups. If they have the same type, then they are isomorphic.*

Corollary 5.3.— *Let S be a scheme and G, G' two splittable S -groups. The following are equivalent:*

- (i) G and G' are isomorphic;
- (ii) they are locally isomorphic for the fpqc topology;
- (iii) there is $s \in S$ such that G_s and G'_s have the same type.

Clearly (i) implies (ii), which implies (iii). Conversely, (iii) gives

$$R(G) = R(G_s) = R(G'_s) = R(G'),$$

so Corollary 5.2 applies.

Corollary 5.4 (Uniqueness of Chevalley schemes).— *Let G, G' be reductive groups over $\mathbf{Z}$, each possessing a split maximal torus. The following are equivalent:*

- (i) G and G' are isomorphic;
- (ii) there is $s \in \text{Spec } \mathbf{Z}$ such that G_s and G'_s have the same type;
- (iii) $G_{\mathbf{C}}$ and $G'_{\mathbf{C}}$ have the same type.

Indeed, G and G' are splittable by Exposé XXII, 2.2. ¹⁰

Corollary 5.5 (Existence of outer automorphisms).— *Let S be a scheme, G a pinned S -group, and h an automorphism of the pinned root datum $R(G)$. There is a unique automorphism u of G preserving the pinning and satisfying $R(u) = h$.*

Corollary 5.5 bis.— *Let S be a scheme, G a split reductive S -group, and R^+ a system of positive roots. For every simple root α , choose an isomorphism of vector groups*

$$p_\alpha : \mathbf{G}_{\alpha, S} \xrightarrow{\sim} U_\alpha.$$

¹⁰*Editors' note.* In fact every $\mathbf{Z}$ -torus is split: every $\mathbf{Z}$ -torus is isotrivial, and every étale cover of $\text{Spec } \mathbf{Z}$ is trivial.

Let h be an automorphism of M permuting the positive roots and the corresponding coroots: if $\alpha \in R^+$, then

$$h(\alpha) \in R^+, \quad h^\vee(\alpha^*) = h(\alpha)^*.$$

There is a unique automorphism u of G inducing $D_S(h)$ on T and satisfying $u \circ p_\alpha = p_{h(\alpha)}$ for every simple root α .

Corollary 5.6.— *Let G, G' be reductive S -groups. The following are equivalent:*

- (i) G, G' are locally isomorphic for the fpqc topology;
- (i bis) they are locally isomorphic for the étale topology;
- (ii) G_s, G'_s are isomorphic for every $s \in S$;
- (iii) the functions $s \mapsto \text{type}(G_s)$ and $s \mapsto \text{type}(G'_s)$ are equal.

The implication (i bis) $\Rightarrow$ (i) is immediate; (i) $\Rightarrow$ (ii) follows from the principle of finite extension (EGA IV₃, 9.1.4); and (ii) $\Rightarrow$ (iii) is immediate. For (iii) $\Rightarrow$ (i bis), one may suppose G, G' splittable (Exposé XXII, 2.3), and then apply Corollary 5.3.

Corollary 5.7.— *Let S be a scheme, G a reductive S -group, and G' an affine smooth S -group with connected fibers. If $s \in S$ and $G_s \simeq G'_s$, then there is an étale morphism $S' \rightarrow S$, covering s , such that $G_{S'} \simeq G'_{S'}$.*

By Exposés XIX, 2.5 and XXII, 2.1, one may suppose G, G' reductive and splittable, and Corollary 5.3 applies.

Corollary 5.8.— *Let k be a field and G, G' reductive k -groups. The following are equivalent:*

- (i) G, G' have the same type;
- (ii) $G \otimes_k \bar{k}$ and $G' \otimes_k \bar{k}$ are isomorphic;
- (iii) for some finite separable extension K/k , $G \otimes_k K$ and $G' \otimes_k K$ are isomorphic.

Corollary 5.9.— *Let S be a nonempty scheme and R a root datum. The following are equivalent:*

- (i) there is a pinned S -group of type R ;
- (ii) there is an S -group of type R ;
- (iii) fpqc-locally, there is a reductive S -group of type R .

Only (iii) $\Rightarrow$ (i) needs proof. Suppose, for simplicity, that there are a faithfully flat quasi-compact morphism $S' \rightarrow S$ and a reductive S' -group G' of type R . We may suppose G' splittable and choose a pinning E' , writing $R = R(G', E')$. The two pullbacks of (G', E') to

$$S'' = S' \times_S S'$$

are pinned groups $(G''_1, E''_1), (G''_2, E''_2)$, with canonical isomorphisms

$$p_i : R(G''_i, E''_i) \xrightarrow{\sim} R.$$

Thus

$$p = p_2^{-1} \circ p_1 : R(G''_1, E''_1) \xrightarrow{\sim} R(G''_2, E''_2).$$

The uniqueness theorem gives a unique isomorphism

$$f : (G''_1, E''_1) \xrightarrow{\sim} (G''_2, E''_2)$$

with $R(f) = p$. This is a descent datum on G' . Its cocycle condition may be checked after applying R , since R is fully faithful, and then follows from the construction of p . Because G' is affine, the datum is effective. The pinning is stable under descent, so it descends to a pinned S -group (G, E) of type R .

N.B. In the language of fibered categories, the preceding proof becomes both shorter and more transparent.

Corollary 5.10.— *Let S be a nonempty scheme and R a pinned root datum for which there exists a reductive S -group of type R . Then there is a pinned S -group of type R , unique up to unique isomorphism.*

Definition 5.11.— *Under these hypotheses, denote the unique pinned S -group of type R by $\acute{E}p_S(R)$, its canonical maximal torus by $T_S(R)$, its canonical Borel subgroup by $B_S(R)$, and so on.*

For every morphism $S' \rightarrow S$, with $S' \neq \emptyset$, one may identify

$$\acute{E}p_{S'}(R) = \acute{E}p_S(R) \times_S S'.$$

If $\acute{E}p_{\text{Spec } \mathbf{Z}}(R)$ exists, write it $\acute{E}p(R)$; then

$$\acute{E}p_S(R) = \acute{E}p(R) \times_{\text{Spec } \mathbf{Z}} S.$$

The group $\acute{E}p(R)$ is called the *Chevalley group scheme of type R* .

5.12.— Thus an S -sheaf in groups G is a reductive S -group of type R if and only if it is locally isomorphic, for the étale or fpqc topology, to $\acute{E}p_S(R)$. Similarly, by the conjugacy theorems, giving a reductive S -group of type R with a maximal torus is locally equivalent to

$$(\acute{E}p_S(R), T_S(R)).$$

The analogous statements hold for Borel subgroups and Killing pairs.

6. Chevalley Systems

The explicit calculations of Section 3 have important numerical consequences.

Definition 6.1.— *Let S be a scheme and (G, T, M, R) a split S -group. A Chevalley system of G is a family*

$$(X_\alpha)_{\alpha \in R}, \quad X_\alpha \in \Gamma(S, \mathfrak{g}^\alpha)^\times,$$

satisfying:

(SC) *for every pair $\alpha, \beta \in R$,*

$$\text{Ad}(w_\alpha(X_\alpha))X_\beta = \pm X_{s_\alpha(\beta)}.$$

Recall that

$$w_\alpha(X_\alpha) = \exp(X_\alpha) \exp(-X_\alpha^{-1}) \exp(X_\alpha).$$

Condition (SC) implies, in particular,

$$X_{-\alpha} = \pm X_\alpha$$

by the identity

$$\text{Ad}(w_\alpha(X_\alpha))X_\alpha = -X_\alpha^{-1}.$$

Proposition 6.2.— *Every split group admits a Chevalley system. More precisely, if*

$$(\Delta, (X'_\alpha)_{\alpha \in \Delta})$$

is a pinning of (G, T, M, R) , there is a Chevalley system $(X_\alpha)_{\alpha \in R}$ with $X_\alpha = X'_\alpha$ for $\alpha \in \Delta$.

First, it suffices to verify (SC) for $\alpha \in \Delta$. Indeed, for every $\alpha \in R$, choose a sequence $\alpha_i \in \Delta$ such that

$$\alpha = s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}).$$

Repeated use of (SC) gives

$$X_\alpha = \pm \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n})) X_{\alpha_{n+1}}.$$

By Lemma 3.1.1(iii),

$$\begin{aligned} w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n}) w_{\alpha_{n+1}}(X_{\alpha_{n+1}}) w_{\alpha_n}(X_{\alpha_n})^{-1} \cdots w_{\alpha_1}(X_{\alpha_1})^{-1} \\ = \alpha^*(\pm 1) w_\alpha(X_\alpha). \end{aligned}$$

Now

$$w_{\alpha_i}(X_{\alpha_i})^{-1} = \alpha_i^*(-1) w_{\alpha_i}(X_{\alpha_i}),$$

and

$$\beta(\gamma^*(-1)) = (-1)^{(\gamma^*, \beta)} = \pm 1.$$

It follows that (SC) for every pair (α_i, γ) implies (SC) for every (α, β) .

For each $\alpha \in R$, choose such a sequence and define

$$X_\alpha = \text{Ad}(w_{\alpha_1}(X_{\alpha_1}) \cdots w_{\alpha_n}(X_{\alpha_n})) X'_{\alpha_{n+1}}.$$

It remains to prove the following lemma.

Lemma 6.3.— *Let*

$$(G, T, M, R, \Delta, (X_\alpha)_{\alpha \in \Delta})$$

be a pinned S -group. If α_i , $0 \leq i \leq n+1$, are simple roots such that

$$s_{\alpha_1} \cdots s_{\alpha_n}(\alpha_{n+1}) = \alpha_0,$$

then

$$\text{Ad}(w_{\alpha_1} \cdots w_{\alpha_n}) X_{\alpha_{n+1}} = \pm X_{\alpha_0}.$$

Arguing as in 2.3.4, using Tits's lemma, reduces the proof to:

(a) G has semisimple rank at most 2; or

(b) $w_{\alpha_1} \cdots w_{\alpha_n}$ is a section of T .

Case (a) is exactly the calculation in 3.1.2(i), 3.2.1(i), 3.3.1(i), or 3.4.1(i).

In case (b), it suffices to show that if

$$s_{\alpha_1} \cdots s_{\alpha_n} = 1,$$

then $t = w_{\alpha_1} \cdots w_{\alpha_n}$ satisfies $\alpha(t) = \pm 1$ for every root α . By the structure of the Weyl group, the word in the free group generated by the s_α lies in the normal subgroup generated by

$$(s_\alpha s_\beta)^{n_{\alpha\beta}}.$$

Thus one reduces to a word

$$s_{\alpha_1} \cdots s_{\alpha_i} (s_{\alpha_{i+1}} s_{\alpha_{i+2}})^{n_{\alpha_{i+1}\alpha_{i+2}}} s_{\alpha_i} \cdots s_{\alpha_1},$$

for which

$$t = s_{\alpha_1} \cdots s_{\alpha_i} (t_{\alpha_{i+1}\alpha_{i+2}}).$$

The assertion is now immediate from the values of $t_{\alpha_1\alpha_2}$ computed in 3.1.2(i), 3.2.1(i), 3.3.1(i), and 3.4.1(i).

Proposition 6.4.— *Let (G, T, M, R) be a split S -group with Chevalley system $(X_\alpha)_{\alpha \in R}$. Let α, β be nonproportional roots, with*

$$\text{length}(\alpha) \leq \text{length}(\beta), \quad |(\beta^*, \alpha)| \leq |(\alpha^*, \beta)|.$$

Let q and $p - 1$ be the nonnegative integers for which the roots of the form $\beta + k\alpha$ are

$$\{\beta - (p - 1)\alpha, \dots, \beta, \dots, \beta + q\alpha\}.$$

Thus $-(\alpha^, \beta) = q - p + 1$. Writing $p_\gamma(x) = \exp(xX_\gamma)$, the commutation relation between U_α and U_β is given by:*

(p, q)	$p_\beta(y)p_\alpha(x)p_\beta(-y)p_\alpha(-x)$
$(-, 0)$	e
$(1, 1)$	$p_{\alpha+\beta}(\pm xy)$
$(1, 2)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)$
$(1, 3)$	$p_{\alpha+\beta}(\pm xy)p_{2\alpha+\beta}(\pm x^2y)p_{3\alpha+\beta}(\pm x^3y)p_{3\alpha+2\beta}(\pm x^3y^2)$
$(2, 1)$	$p_{\alpha+\beta}(\pm 2xy)$
$(2, 2)$	$p_{\alpha+\beta}(\pm 2xy)p_{2\alpha+\beta}(\pm 3x^2y)p_{\alpha+2\beta}(\pm 3xy^2)$
$(3, 1)$	$p_{\alpha+\beta}(\pm 3xy)$

These are all possibilities, since the root string has length at most 3.

Proof. By Exposé XXI, 3.5.4, there is a system of simple roots Δ such that $\alpha \in \Delta$, and there are $\alpha' \in \Delta$, $a, b \in \mathbf{Q}_+$, with

$$\beta = a\alpha + b\alpha'.$$

The desired relation lies in $U_{\alpha\alpha'}$, so it follows from the rank-two calculations of Section 3 and condition (SC).

Corollary 6.5 (Chevalley's rule).— *Let $(X_\alpha)_{\alpha \in R}$ be a Chevalley system of the split S -group G . If $\alpha, \beta, \alpha + \beta \in R$, then*

$$[X_\alpha, X_\beta] = \pm pX_{\alpha+\beta},$$

where p is the least positive integer for which $\beta - p\alpha$ is not a root.

The assertion is symmetric in α, β ; after arranging $\text{length}(\alpha) \leq \text{length}(\beta)$, it follows from Proposition 6.4.

Corollary 6.6.— *Let S be a scheme such that $6 \cdot 1_S \neq 0$, and let (G, T, M, R) be a split S -group. If $R' \subset R$ is such that*

$$(*) \quad \mathfrak{g}_{R'} = \mathfrak{t} \oplus \bigoplus_{\alpha \in R'} \mathfrak{g}^\alpha$$

is a Lie subalgebra of $\mathfrak{g}$, then R' is a closed subset of R .

Choose a Chevalley system and let $\alpha, \beta \in R'$ with $\alpha + \beta \in R$. By Corollary 6.5 and Proposition 6.4,

$$[X_\alpha, X_\beta] = \pm p X_{\alpha+\beta}, \quad p \in \{1, 2, 3\}.$$

Since neither 2 nor 3 is zero on S , $(*)$ implies $\mathfrak{g}^{\alpha+\beta} \subset \mathfrak{g}_{R'}$, and hence $\alpha + \beta \in R'$.¹¹

6.7.— The exact signs in this section can be determined by studying $\text{Norm}_G(T)$, and more precisely the extended Weyl group

$$W^* = N(\mathbf{Z}), \quad N = \text{Norm}_{\hat{E}_{\mathbf{P}\mathbf{Z}}(R)}(T_{\mathbf{Z}}(R)).$$

It is an extension of $W(R)$ by an abelian group of type $(2, 2, \dots, 2)$, which accounts for the signs.

Remark 6.7.1.— *By 3.1.2(i) and 3.n.1, $n = 2, 3, 4$, the elements $w_\alpha, w_\beta \in N(\mathbf{Z})$ satisfy the braid relations*

$$w_\alpha w_\beta \cdots = w_\beta w_\alpha \cdots$$

with $n_{\alpha\beta}$ factors on each side.

See also Tits [Ti66], Bourbaki [BLie], IX.4, Ex. 12, and Springer [Sp98, 9.3.2].

¹¹*Editors' note.* It suffices that 2 and 3 are nonzero on S ; for example the result applies to $S = \text{Spec}(\mathbf{Z}/6\mathbf{Z})$. In the excluded characteristics, short-root subsystems furnish standard counterexamples.

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